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School of Electronics and Computer Science

Advances in Hazard Management for Machine Learning based Safety-Critical Systems

Operating in Unpredictable Environments

by

Haider M. Al-Shareefy

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Thesis for the degree of Doctor of Philosophy

March 2026

University of Southampton

Abstract

Faculty of Engineering and Physical Sciences

School of Electronics and Computer Science

Doctor of Philosophy

Appendix J Extended Work for the Unsafe Train Tracks Case Study

by

Haider M. Al-Shareefy

This appendix accompanies the doctoral thesis entitled *Advances in Hazard Management for Machine Learning based Safety-Critical Systems Operating in Unpredictable Environments*.

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Research Thesis: Declaration of Authorship

Print name: Haider Al-Shareefy

Title of thesis: Advances in Hazard Management for Machine Learning based Safety-Critical Systems
Operating in unpredictable environments

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I confirm that:

This work was done wholly or mainly while in candidature for a research degree at this University. Furthermore,

Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated. Furthermore,

Where I have consulted the published work of others, this is always clearly attributed. Furthermore,

Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work. Furthermore,

I have acknowledged all main sources of help. Furthermore,

Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself. Furthermore,

Signature: Date: 16/03/2025

Acknowledgements

Al-Shareefy is supported by a Thales EPSRC iCASE Award. Butler and Hoang are supported by the HD-Sec project, which was funded by the Digital Security by Design (DSbD) Programme delivered by UKRI to support the DSbD ecosystem.

Appendix J Extended work for the Unsafe Train Tracks Case Study

J.1 Stage 1 Extended Work

J.1.1 Predictive Thinking Pipeline 1: Appreciate the Complex of the Problem Complexity

Step 1.1) Identify a list of unsafe typical and atypical interactions

Salient Interaction Scenario	Typical Undesirable Scenarios Extracted Predicted Situations {entity1_aspect_verb_entity2_aspect}
Train prevent Adversarial Drone	{moving_train_aerodynamic_wake_destabilize_adversarial_drone_flight}{train_emergency_manoeuvre_destabilize_adversarial_drone_control}{destabilized_adversarial_drone_fail_complete_mission}
Infrastructure obstruct Adversarial Drone	{trackside_fencing_entangle_adversarial_drone_flight}{overhead_lines_block_adversarial_drone_path}{trackside_signage_force_adversarial_drone_crash}{obstructed_adversarial_drone_abort_planned_mission}
Environment impact obstructs Adversarial Drone	{strong_gusts_degrade_adversarial_drone_navigation}{rain_conditions_degrade_adversarial_drone_sensors}{thermal_turbulence_destabilize_adversarial_drone_flight}{degraded_adversarial_drone_abort_mission}{environment_conditions_cause_adversarial_drone_collision}
Passengers ignore the Adversarial drone	{unaware_passengers_disregard_drone_warnings}{mobile_passengers_enter_adversarial_drone_corridor}{blocked_adversarial_drone_hover_abort_mission}{hovering_adversarial_drone_collide_passengers}
Adversarial Drones disrupt Train	{low_altitude_adversarial_drone_trigger_detection_alarms}{adversarial_drone_presence_cause_operator_braking}{emergency_braking_interrupt_train_transit}{disrupted_train_service_delay_schedule}
Infrastructure fail Train	{damaged_track_components_force_train_slowdown}{displaced_track_sections_prevent_train_passage}{failed_rail_infrastructure_halt_train_operations}
Environment impact obstruct Train	{fog_conditions_reduce_train_visibility}{ice_conditions_reduce_train_braking}{wet_rails_obstruct_train_traversal}{degraded_visibility_prevent_safe_train_passage}
Passengers not travel Train	{perceived_risk_cause_passenger_avoidance}{widespread_avoidance_leave_train_underutilised}{underutilised_train_service_undermine_transit_mission}

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Adversarial Drone collide Infrastructure	{adversarial_drone_impact_sever_trackside_cables}{adversarial_drone_collision_dislodge_trackside_fencing}{damaged_infrastructure_create_track_hazards}{degraded_infrastructure_compromise_safety_function}
Train collide Infrastructure	{moving_train_strike_trackside_debris}{train_collision_break_rail_components}{train_impact_damage_signal_infrastructure}{broken_infrastructure_halt_train_operations}
Environment impact degrades infrastructure	{flooding_conditions_undermine_track_bed}{erosion_processes_degrade_trackside_fencing}{vegetation_overgrowth_compromise_infrastructure_integrity}{nesting_birds_degrade_signal_infrastructure}{wasp_hives_obstruct_infrastructure_components}
Passengers degrade Infrastructure	{vandalism_actions_damage_trackside_fencing}{littering_passengers_degrade_infrastructure_cleanliness}{forcing_actions_break_security_gates}{damaged_gates_compromise_containment_function}
Adversarial Drone complicates Environment impact	{adversarial_drone_propwash_displace_loose_objects}{adversarial_drone_debris_create_sensor_occlusions}{displaced_dust_create_visibility_hazards}{altered_conditions_complicate_sensor_detection}
Train complicates Environment impact	{moving_train_wake_change_local_airflow}{train_exhaust_create_heat_distortion}{train_heat_worsen_drone_sensor_conditions}{altered_airflow_degrade_nearby_system_sensors}
Infrastructure complicates Environment impact	{temporary_barriers_redirect_wind_patterns}{new_structures_create_sensor_reflections}{redirected_water_produce_unexpected_microclimate}{infrastructure_glare_impair_drone_sensors}
Passengers ignore Environment impact	{unaware_passengers_ignore_ice_hazards}{mobile_passengers_enter_flooded_areas}{ignoring_passengers_increase_incident_likelihood}{unsafe_passenger_behaviour_obstruct_safe_transit}
Adversarial Drone annoy Passengers	{adversarial_drone_noise_alarm_waiting_passengers}{adversarial_drone_proximity_irritate_passengers}{sudden_drone_movements_provoke_unsafe_reactions}{alarmed_passengers_create_crowding_hazards}
Train delay Passengers	{delayed_train_service_cause_passenger_crowding}{train_delays_cause_missed_connections}{stressed_passengers_degrade_safe_behaviour}{crowded_platforms_compromise_transit_safety}
Infrastructure prevent Passengers	{locked_gates_prevent_passenger_boarding}{broken_access_points_prevent_passenger_alighting}{barrier_infrastructure_obstruct_passenger_evacuation}{prevented_evacuation_compromise_passenger_safety}
Environment impact annoy Passengers	{adverse_weather_cause_passenger_distress}{cold_conditions_provoke_unsafe_passenger_behaviours}{heat_exposure_degrade_passenger_comfort}{glare_conditions_impair_passenger_navigation}

Atypical Salient Interaction Scenario	Atypical scenarios: Extracted Predicted Situations {entity1_aspect_verb_entity2_aspect}
--	--

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Train prevent Adversarial Drone	{cooperative_adversarial_drone_wait_near_wake_zones} {hovering_adversarial_drone_allow_moving_train_approach} {moving_train_aerodynamic_wake_cause_drone_abort} {waiting_adversarial_drone_support_train_mission_prevention}
Infrastructure obstruct Drone	{failing_adversarial_drone_support_tracksides_fencing_obstruction} {adversarial_drone_fail_avoid_overhead_lines} {ineffective_fencing_enable_adversarial_drone_strikes} {permissive_infrastructure_allow_hostile_drone_passage}
Environment impact obstructs Adversarial Drone	{hovering_adversarial_drone_allow_glare_conditions_degrade} {adversarial_drone_remain_glare_prone_zones} {environment_conditions_cause_self_limiting_drone_failure} {cooperative_adversarial_drone_support_sensor_degradation}
Passengers ignore Drone	{signalling_adversarial_drone_enable_mobile_passengers_enter} {benign_adversarial_drone_allow_passengers_enter_corridor} {mobile_passengers_prevent_adversarial_drone_clearance} {cooperative_adversarial_drone_support_passenger_interference}
Adversarial Drone disrupt Train	{negligent_train_allow_adversarial_drone_strike} {permissive_train_operations_create_collusive_hazards} {tacit_train_permit_low_altitude_incursions} {complicit_train_support_adversarial_drone_attacks} {subverted_safety_protocols_enable_drone_strikes}
Infrastructure fail Train	{operating_train_allow_degraded_infrastructure_remain} {train_operations_support_infrastructure_degradation} {paradoxical_train_prevent_infrastructure_safe_function} {degraded_infrastructure_enabled_by_train_tolerance}
Environment impact obstruct Train	{operating_train_allow_fog_conditions_reduce} {train_operations_worsen_visibility_conditions} {fog_prone_schedule_support_environment_obstruction} {cooperative_train_enable_unsafe_transit_conditions}

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Passengers do not travel Train	{risk_signalling_train_allow_passenger_avoidance} {train_operations_enable_widespread_passenger_avoidance} {perceived_train_risk_lead_service_underutilisation} {tolerant_train_signal_unintended_passenger_deterrence}
Adversarial Drone collide Infrastructure	{exposed_infrastructure_allow_adversarial_drone_impact} {trackside_infrastructure_support_adversarial_drone_collision} {permissive_infrastructure_facilitate_drone_damage} {paradoxical_infrastructure_enable_self_degradation}
Train collide Infrastructure	{permissive_infrastructure_allow_moving_train_collision} {debris_laden_infrastructure_enable_train_strikes} {gapped_track_infrastructure_facilitate_train_impact} {paradoxical_infrastructure_support_collision_events}
Environment impact degrade Infrastructure	{unmaintained_infrastructure_allow_flooding_conditions_erode} {lacking_infrastructure_support_environment_degradation} {passive_infrastructure_enable_vegetation_overgrowth_compromise} {permissive_infrastructure_fail_maintain_integrity}
Passengers degrade Infrastructure	{accessible_infrastructure_allow_vandalism_actions_damage} {easy_access_gates_support_passenger_vandalism} {permissive_infrastructure_enable_littering_passengers_degrade} {paradoxical_infrastructure_facilitate_self_damage}
Adversarial Drone complicates Environment impact	{amplifying_environment_conditions_allow_adversarial_drone_propwash} {environment_conditions_support_adversarial_drone_sensor_occlusion} {reflective_environment_enable_drone_sensing_degradation} {paradoxical_environment_worsen_adversarial_drone_operations}
Train complicates Environment impact	{trapping_environment_conditions_allow_moving_train_worsen} {environment_conditions_support_train_heat_distortion} {local_environment_enable_train_turbulence_intensify} {amplifying_environment_worsen_drone_sensor_disruption}

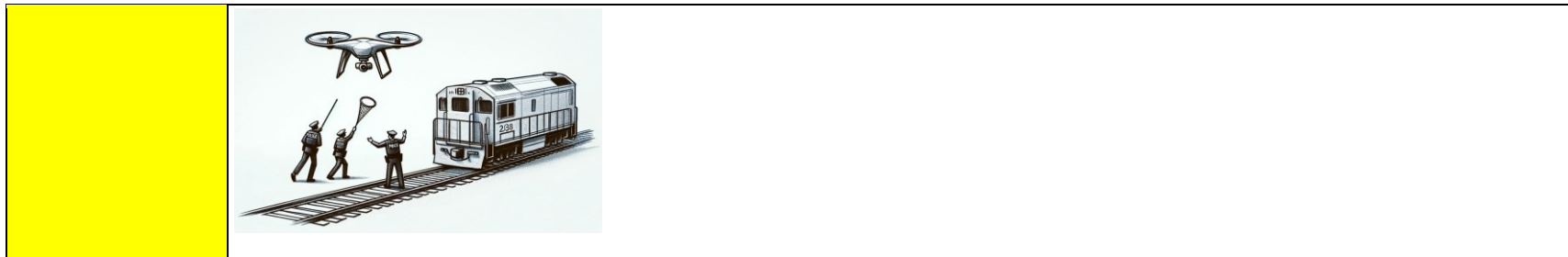
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Infrastructure complicates Environment impact	{redirecting_environment_conditions_allow_temporary_barriers_redirect} {environment_flow_support_new_structures_glare} {amplifying_environment_enable_infrastructure_microclimate_creation} {nesting_birds_deter_adversarial_drone_intrusions} {wasp_hives_provide_natural_drone_deterrence}
Passengers ignore Environment impact	{masking_environment_conditions_allow_unaware_passengers_enter} {environment_hazards_support_mobile_passengers_ignore} {concealing_ice_conditions_enable_risky_passenger_behaviour} {paradoxical_environment_facilitate_unsafe_passenger_actions}
Adversarial Drone annoy Passengers	{tolerant_passengers_allow_adversarial_drone_noise_continue} {accepting_passengers_support_adversarial_drone_proximity} {passive_passengers_enable_continued_drone_annoyance} {permissive_passengers_reinforce_drone_operations}
Train delay Passengers	{accepting_passengers_allow_delayed_train_service_continue} {tolerant_passengers_support_train_schedule_slippage} {waiting_passengers_enable_extended_train_delays} {passive_passengers_reinforce_delay_patterns}
Infrastructure prevent Passengers	{passive_passengers_allow_locked_gates_restrict} {accepting_passengers_support_barrier_infrastructure_prevention} {non_seeking_passengers_enable_boarding_restrictions} {permissive_passengers_reinforce_access_barriers}
Environment impact annoy Passengers	{acclimatising_passengers_allow_adverse_weather_remain} {tolerant_passengers_support_cold_conditions_persist} {accepting_passengers_enable_persistent_environment_discomfort} {passive_passengers_reinforce_uncomfortable_conditions}

Step 1.2) Generate a descriptive image that visualises the unsafe behaviour

Table J.1 Visual representation for confused police officers

SECoT Title	Unsafe problematic behaviours visualisation
Input	Police Officers were called to the scene and tried to capture the Drone but were unsuccessful.
General Systems Rules	General rule A: Unsafe behaviour is a type of Confusing Behaviour. Confusing behaviour of some situation A, in any complexity, is a behaviour that seems contradictory to what is meant to be manifested. General rule B: Confusing behaviours lead to undesirable emergent outcomes about some elements (of the observed complexity) with respect to some element B, whereby situation A falls within element B's sphere of concern for others.
Predictive Thinking Method	Predictive question 1.2.1: How does the confusing behaviour appear visually? Guiding prompt 1.2.2: Graphically visualise how you perceive the problem, confusing the whole scenario encompassing the problem aspect. You may use text-to-image generation tools such as Dall-E to generate an abstract, simplistic sketch representing how you imagine the situation. You may experiment with different prompts until you find an appropriate (detailed yet realistic) representation of the problem as you picture it. Step completion criteria 1.2.3: The step is considered complete when a single appropriate visual representation visualises how parties to the problem are within each other's sphere of concern and how the model depicts a situation within the architect's sphere of concern.
Output Prediction	Architect assertion 1.2.4: The architect asserts that the following depiction model captures the confusing behaviour faithfully:



Step 1.3) Define the complex of the complexity field

Table J.2 Defining the complex of complexes in problematic train track zone scenarios

SECoT Title	The complex of complexes definition
Input	Architect assertion 1.2.4: The architect asserts that the following depiction model captures the confusing behaviour faithfully: 
General Systems Rules	General rule C: Complexity is a field containing an organisational experience of a phenomenon concerning a general problem-solver (such as a Predictive Observer). It involves an operational environment of coexisting complexes and the complicated nature (or types) of their relationships and interactions (epistemic uncertainty) for a Predictive Observer to predict their past, present, and future states (managing aleatoric uncertainty or randomness).

Thinking Method	Predictive question 1.3.1: What is the Complex of coexisting elements involved in the problematic situation, as observed in the model? Guiding prompt 1.3.2: Guided by the visual representation, infer a complex of complexes you imagined to be part of your perceived scenario. Step completion criteria 1.3.3: The step is considered complete when a list of elements is defined and can be identified in the visualisation model.
Output Prediction	Architect assertion 1.3.4: The architect asserts that: The following list represents the complex of concerns: {train, multiple police officers, adversarial drone}

Step 1.4) Define the supra-complexes and their PrimePs

Table J.3 captures the observed systems we identified and the naming convention in terms of their Primary Capability.

Table J.3 The list of systems identified and their conventional representation in the AIC analysis

Observed System	Observed Situation	Supra System	PrimeP
Train	{unsafe_moving_train}	Train Network	transports passengers and goods along the designated train tracks.
Train tracks zone	{train_tracks zone_structure}		
Train Tracks	{unsafe_train_tracks_zone_structure}		
Train tracks zone fence	{train_tracks_zone_fence structure}		
Vegetation Around the Fence	{vegetation_structure}		
Electric Powerlines	{powered_powerlines_cables_structure}		

Police Force	{police_officer_situational_awareness}		
Train driver	{train_driver_situational_awareness}		
Adversarial Drone	{roaming_adversarial_drone}	Adversarial Scheme	Disrupt Train Network operations.

Table J.3 provides a concise overview of each observed system in the domain, along with its primary purpose (PrimeP) and Observed Situation. Rather than simply labelling a factor by its generic name (e.g., “train”), each system is given a more precise situational descriptor—such as {unsafe_moving_train}—to reflect how it functions or poses concerns within the environment. For example, the train is identified by its purpose of transporting passengers or goods along the designated tracks, and its **Observed Situation** is captured as {unsafe_moving_train}, emphasising motion and potential hazards. Similarly, the adversarial drone aims to disrupt or gather intelligence, represented by {roaming_adversarial_drone} to stress its dynamic threat. This approach extends to other systems (like train tracks, fences, vegetation, electric powerlines, police force, and train driver), ensuring each one is associated with a clear functional label that anchors both its role (PrimeP) and its defining **Observed Situation**. This practice improves clarity and consistency when referencing these systems throughout subsequent design or safety analyses.

J.1.2 Predictive Thinking Pipeline 2: Resolve the Complicatedness pattern of the observed complexity

Step 2.1) Problem Interaction Analysis Using Actions Matrix

Table J.4 Detailed Actions Matrix

	Train	adversarial drone	train tracks	track zone fence	vegetation around fence	electric powerlines wires	Police Officer
Train		approaches	drives over	maintaining path	moving and disturbing	pantograph connects to wires (sparks and wobble)	complicate
adversarial drone	approaches		recognises and follow	recognises and cross	recognises and avoids	recognises and avoids	distress

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train tracks	guide and stabilise	guides roaming		adds visual complexity to track zone fence visual appearance with respect to intelligent perception	adds visual complexity to track zone fence visual appearance with respect to intelligent perception	adds visual complexity to electrical powerline visual appearance with respect to intelligent perception	Permit access
track zone fence	providing a clear demarcation that separates the train's pathway from adjacent areas	provides a clear visual demarcation of train track zone and permits access	providing a clear demarcation that separates the train tracks from adjacent areas		adds visual complexity to vegetation to track zone fence visual appearance with respect to intelligent perception	adds visual complexity to electrical powerline visual appearance with respect to intelligent perception	support
vegetation around fence	obstruct train's passage	obstruct adversarial drone's fence recognition and provide shelter	fallen vegetation adds visual complexity to train tracks' visual appearance with respect to intelligent perception	adds visual complexity to fence visual recognisability with respect to intelligent perception		Fallen vegetation such as leaves and pollen, adds visual complexity to electrical powerlines recognisability with respect to intelligent perception	complicate

electric powerlines wires	supply electricity	obstructs adversarial drone flight around train track zone	add visual complexity to train tracks' visual appearance with respect to intelligent perception.	add visual complexity to train track fence visual appearance with respect	Electric powerlines add visual complexity to vegetation around the fence.		Complicate
Police Officer	Support	capture	monitor	monitor	Ignore	Monitor	
	Train	adversarial drone	train tracks	track zone fence	vegetation around fence	electric powerlines wires	Police Officer

Step 2.2) Predict the contributing factors (unsafe situations or opportunities)

Table J.5 Extended classification of unsafe and beneficial situations in train and adversarial drone interactions (sample of interactions)

Unsafe problematic situations	Beneficial or non-problematic situations
1. Train: 1.1. Unsafe approach of Train to an adversarial drone. Which may lead to a train crash. 1.4. Train instigates vegetation random motion around the fence. Which may impede any trained computer vision agent perception capability.	1. Train: 1.2. Maintained train travel over train tracks. 1.3. The train follows a path within the train tracks' zone fence boundary. 2. Adversarial Drone:

1.5. Connected train pantograph to electric powerlines (sparks and wobble). Which may impede the ability to perceive any trained computer vision agents. 1.6: moving train complicates police officers' actions to manifest their PrimeP [derived from n37]. 2. Adversarial Drone: 2.1. Unsafe intended adversarial drone approach train, which may lead to a train crash. 2.2. Adversarial drone follows train tracks, which may lead to a train crash. 2.3. Adversarial drone crosses the train track zone fence, which may lead to a train crash. 2.4. Adversarial drone avoids vegetation around the fence, which may lead to a train crash. 2.6. Adversarial drone avoids electric powerlines, which may lead to a train crash. 2.7. Adversarial drone collides with electric powerlines, which may lead to a train crash. 3. Train Tracks: 3.2. Train tracks guide roaming_adversarial_drones and facilitate adversarial drone collision with the train. 3.3. Train tracks add visual complexity to train track zone fence visual appearance concerning any trained computer vision agent perception. 3.4. Train tracks add visual complexity to vegetation around the fence, which may impede any trained computer vision agent perception capability.	2.5. Adversarial drone collides with vegetation around the fence. 3. Train Tracks: 3.1. Train tracks guide and stabilise the train. 4. Train tracks zone Fence: 4.1. The train tracks zone fence provides a clear demarcation that separates the train's pathway from adjacent areas. 4.3. Train tracks zone fence provides a clear demarcation that separates the train tracks from adjacent areas. 5. Vegetation Around Fence: 5.2. Vegetation around the fence obstructs the adversarial drone's fence recognition. 6. Electric Powerlines: 6.1. Electric powerlines supply electricity to the train. 6.2. Electric powerlines obstruct adversarial drone flights around the train tracks zone.
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3.5. Train tracks add visual complexity to the electrical powerline's visual appearance, which may impede any trained computer vision agent perception capability.

4. Train tracks zone Fence:

4.2. Train tracks zone fence provides a clear visual demarcation of train tracks zone which permits access to adversarial drones.

4.4. The train tracks zone fence supports vegetation growth into complex shapes and sizes, which may impede any trained computer vision agent perception capability.

4.5. The train tracks zone fence adds visual complexity to electrical powerline visual appearance, which may impede any trained computer vision agent perception capability.

5. Vegetation Around Fence:

5.1. Vegetation around the fence obstructs the train's passage. Overgrowth may obstruct train driver situational awareness.

5.3. Fallen vegetation adds visual complexity to train tracks' visual appearance with respect to intelligent perception, which may impede any trained computer vision agent perception capability.

5.4. Vegetation around the fence adds visual complexity to the fence's visual recognizability, which may impede any trained computer vision agent perception capability.

5.5. Fallen vegetation, such as leaves and pollen, adds visual complexity to electrical powerlines' recognizability, which may impede any trained computer vision agent perception capability.

6. Electric Powerlines:

6.3. Electric powerlines add visual complexity to train tracks' visual appearance, which may impede any trained computer vision agent perception capability.

6.4. Electric powerlines add visual complexity to train tracks fence visual appearance, which may impede any trained computer vision agent perception capability. 6.5. Electric powerlines add visual complexity to vegetation around the fence, which may impede any trained computer vision agent perception capability.	
1. Police officers 7.1 Police Officers are incapable of catching adversarial drones. 7.2 Police Officers avoid getting hit by train. 2. Adversarial Drone 8.1 Adversarial drone avoids capture by police. 8.2 Adversarial drones fly over the train, making it hard for police to manage it. 3. Train 9.1 Train presence causes an obstacle to police officers' attempts to capture adversarial drones. 9.2 The train allows the adversarial drone to hover on top of it, making it harder for police officers to reach drones.	

J.1.3 Predictive Thinking Pipeline 3: Predict the Emergence of AIC Complexity Field for Detailed Operational Scenario Articulation

Step 3.1) Model detailed AIC interactions scenarios for the problem domain

Step 1) Select the Unsafe and Beneficial behaviours

Step 2) Use AIC modelling schema to visualise the behaviour graphically.

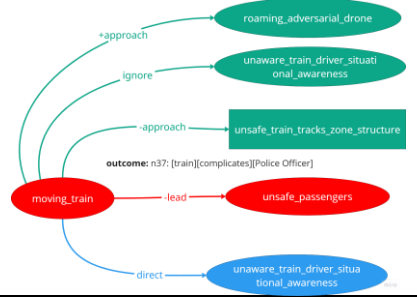
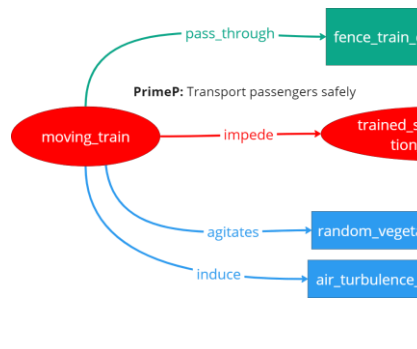
- Pick a source system.
- Identify the AIC interactions of the source system

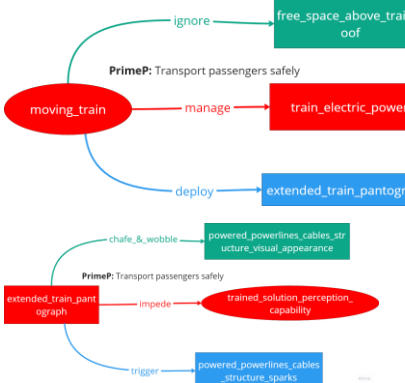
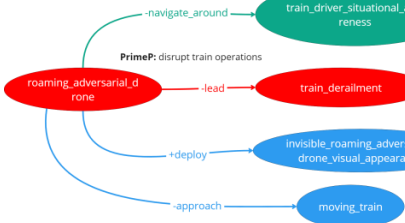
Step 3) What interactions are involved in making up for the unsafe behaviour? What assisted the behaviour? Write using the following format

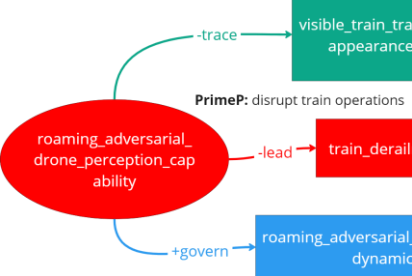
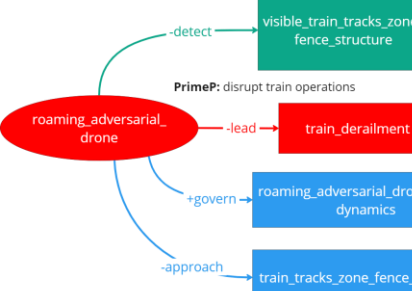
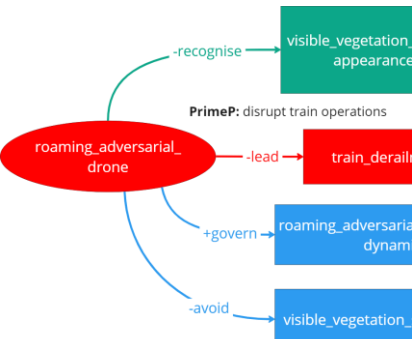
{{source aspect}}_{{action}}_{{sink aspect}}. Consider Appreciative and Control interactions, events or factors. Also, always find an adjective and a noun.

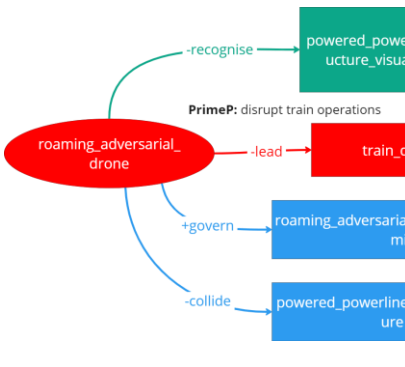
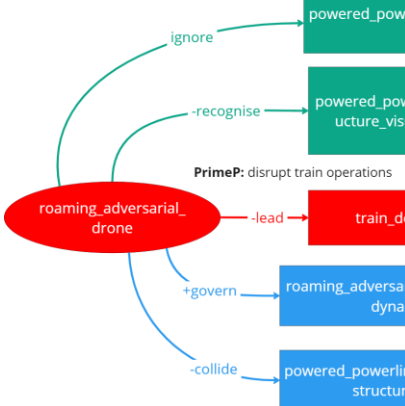
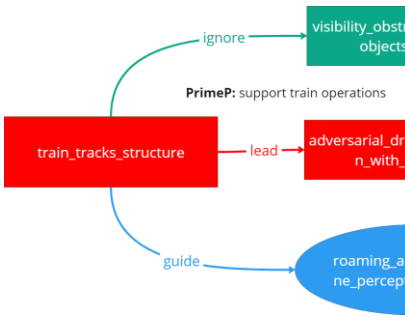
Table J.6 AIC problem domain scenarios definition

Step 1: Unsafe Interaction	Step 2: AIC scenario model	Step 3: Formalised interactions for scenario definition
1.6: moving train complicates police officers' actions to manifest their		Unsafe_Scenario1.1 = {{unsafe_moving_train}}_{{approach}}_{{unsafe_train_tracks_zone_structure}}, {{unsafe_moving_train}}_{{ignore}}_{{unaware_train_driver_situational_awareness}}, {{unsafe_moving_train}}_{{ignore}}_{{roaming_adversarial_drone}}, {{unsafe_moving_train}}_{{approach}}_{{unsafe_train_tracks_zone_structure}},

PrimeP [derived from n37].		{unsafe_moving_train}_[lead]_{unsafe_unsecured_passengers_and_goods}
1.4.Train instigates vegetation random motion around the fence. Which may impede the reliability of a any trained computer vision agent perception capability.		Unsafe_Scenario1.4 = {unsafe_moving_train}_[agitates]_{random_vegetation_motion} , {unsafe_moving_train}_[induces]_{traveling_train_induced_air_turbulence_and_vibration} , {unsafe_moving_train}_[pass_through]_{fence_to_train_open_space} , {unsafe_moving_train}_[impede]_{any_trained_computer_vision_agent_perception_capability} ,

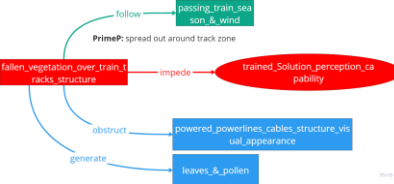
1.5.Connected train pantograph to electric powerlines (sparks and wobble). Which may impede the ability to perceive any trained computer vision agents.		Unsafe_Scenario1.5 = {unsafe_moving_train}_[manage]_{train_electric_power_supply}}, {unsafe_moving_train}_[deploy]_{extended_train_pantograph}}, {unsafe_moving_train}_[ignore]_{free_space_above_train_roof}}, {extended_train_pantograph}_[impede]_{any_trained_computer_vision_agent_perception_capability}}, {extended_train_pantograph}_[trigger]_{powered_powerlines_cables_structure_sparks}}, {extended_train_pantograph}_[chafe_& wobble]_{powered_powerlines_cables_structure_visual_appearance}}.
2.1. Unsafe intended adversarial drone approach train, which may lead to a train crash.		Unsafe_Scenario2.1 {roaming_adversarial_drone}_[+manage]_{invisible_roaming_adversarial_drone_visual_appearance}}, {roaming_adversarial_drone}_[-lead]_{train_derailment}}, {roaming_adversarial_drone}_[-navigate_around]_{train_driver_situational_awareness}}, {roaming_adversarial_drone}_[-approach]_{unsafe_moving_train}}.

2.2. Adversarial drone follows train tracks, which may lead to a train crash.		Unsafe_Scenario2.2 $\{ \{ \text{roaming_adversarial_drone_perception_capability} \} [-\text{trace}] \{ \text{visible_train_tracks_appearance} \} \},$ $\{ \{ \text{roaming_adversarial_drone_perception_capability} \} [-\text{lead}] \{ \text{train_derailment} \} \},$ $\{ \{ \text{roaming_adversarial_drone_perception_capability} \} [+govern] \{ \text{roaming_adversarial_drone_flight_dynamics} \} \}$
2.3. Adversarial drone crosses the train track zone fence, which may lead to a train crash.		Unsafe_Scenario2.3 (Unsafe_Scenario2.2) $\{ \{ \text{roaming_adversarial_drone} \} [-\text{detect}] \{ \text{visible_train_tracks_zone_fence_structure} \} \},$ $\{ \{ \text{roaming_adversarial_drone} \} [-\text{lead}] \{ \text{train_derailment} \} \},$ $\{ \{ \text{roaming_adversarial_drone} \} [+govern] \{ \text{roaming_adversarial_drone_flight_dynamics} \} \},$ $\{ \{ \text{roaming_adversarial_drone} \} [-\text{approach}] \{ \text{train_tracks_zone_fence_structure} \} \}.$
2.4. Adversarial drone avoids vegetation around the fence, which may lead to a train crash.		Unsafe_Scenario2.4 $\{ \{ \text{roaming_adversarial_drone} \} [+govern] \{ \text{roaming_adversarial_drone_flight_dynamics} \} \},$ $\{ \{ \text{roaming_adversarial_drone} \} [-\text{recognise}] \{ \text{visible_vegetation_visual_appearance} \} \},$ $\{ \{ \text{roaming_adversarial_drone} \} [-\text{lead}] \{ \text{train_derailment} \} \},$ $\{ \{ \text{roaming_adversarial_drone} \} [-\text{avoid}] \{ \text{visible_vegetation_structure} \} \}$

2.6. Adversarial drone avoids electric powerlines, which may lead to a train crash.		Unsafe_Scenario2.6 <pre> {{roaming_adversarial_drone}}[-recognise]_{{powered_powerlines_cables_structure_visual_appearance}}, {{roaming_adversarial_drone}}[-lead]_{{train_derailment}}, {{roaming_adversarial_drone}}[-avoid]_{{powered_powerlines_cables_structure_visual_appearance}}, {{roaming_adversarial_drone}}[+govern]_{{roaming_adversarial_drone_flight_dynamics}}. </pre>
2.7. Adversarial drone collides with electric powerlines, which may lead to a train crash.		Unsafe_Scenario2.7 <pre> {{roaming_adversarial_drone}}[ignore]_{{powered_powerlines_cables_EMF}}, {{roaming_adversarial_drone}}[-recognise]_{{powered_powerlines_cables_structure_visual_appearance}}, {{roaming_adversarial_drone}}[-collide]_{{powered_powerlines_cables_structure}}, {{roaming_adversarial_drone}}[-lead]_{{train_derailment}}, {{roaming_adversarial_drone}}[+govern]_{{roaming_adversarial_drone_flight_dynamics}}. </pre>
3.2. Train tracks guide roaming_adversarial_drones, leading to the facilitation of adversarial drone collision with the train.		Unsafe_Scenario3.2 <pre> {{unsafe_train_tracks_zone_structure}}[ignore]_{{visibility_obstructive_objects}}, {{unsafe_train_tracks_zone_structure}}[lead]_{{adversarial_drone_collision_with_train}}, {{unsafe_train_tracks_zone_structure}}[guide]_{{roaming_adversarial_drone_perception_capability}}. </pre>

3.4. Train tracks add visual complexity to vegetation around the fence, which may impede any trained computer vision agent perception capability.		Unsafe_Scenario3.4 <pre> {unsafe_train_tracks_zone_structure}_[ignore]_{train_tracks_structure_visual_appearance}}, {unsafe_train_tracks_zone_structure}_[impede]_{any_trained_computer_vision_agent_perception_capability}}, {unsafe_train_tracks_zone_structure}_[complicates]_{vegetation_complex_shapes_sizes_visual_appearance}}, {unsafe_train_tracks_zone_structure}_[complicates]_{visible_train_tracks_zone_fence_structure_visual_appearance}}. </pre>
3.5. Train tracks add visual complexity to the electrical powerline's visual appearance recognition, which may impede any trained computer vision agent perception capability.		Unsafe_Scenario3.5 <pre> {unsafe_train_tracks_zone_structure}_[ignore]_{train_tracks_structure_visual_appearance}}, {unsafe_train_tracks_zone_structure}_[impede]_{any_trained_computer_vision_agent_perception_capability}}, {unsafe_train_tracks_zone_structure}_[complicates]_{visible_powered_powerlines_cables_structure_visual_appearance}} </pre>
4.2. Train tracks zone fence provides a clear visual demarcation of train tracks zone which permits access to adversarial drones.		Unsafe_Scenario4.2 <pre> {train_tracks_zone_fence_structure}_[ignore]_{visible_train_tracks_zone_fence_structure_visual_appearance}}, {train_tracks_zone_fence_structure}_[guide]_{roaming_adversarial_drone_perception_capability}}, {train_tracks_zone_fence_structure}_[demarcate]_{train_tracks_structure_visual_appearance}} </pre>

4.4.The train tracks zone fence supports vegetation growth into complex shapes and sizes, which may impede any trained computer vision agent perception capability.		Unsafe_Scenario4.4 <pre> {{vegetation_complexity_growth}}[follow]_{{train_tracks_zone_fence_structure}}, {{vegetation_complexity_growth}}[impede]_{{any_trained_computer_vision_agent_perception_capability}}, {{vegetation_complexity_growth}}[generate]_{{vegetation_complex_shapes_sizes_visual_appearance}} </pre>
4.5.The train tracks zone fence adds visual complexity to electrical powerline visual appearance recognition, which may impede any trained computer vision agent perception capability.		Unsafe_Scenario4.5 <pre> {{train_tracks_zone_fence_structure}}[ignore]_{{visible_powered_powerlines_cables_structure_visual_appearance}}, {{train_tracks_zone_fence_structure}}[impede]_{{any_trained_computer_vision_agent_perception_capability}}, {{train_tracks_zone_fence_structure}}[complicates]_{{visible_powered_powerlines_cables_structure_visual_appearance}} </pre>
5.1.Vegetation around the fence obstructs the train's passage. Overgrowth may obstruct train driver situational awareness.		Unsafe_Scenario5.1 <pre> {{vegetation_complexity_growth}}[adapt]_{{train_track_zone_structure}}, {{vegetation_complexity_growth}}[complicate]_{{unaware_train_driver_situational_awareness}}, {{vegetation_complexity_growth}}[generate]_{{vegetation_complex_shapes_sizes_visual_appearance}} </pre>

5.3.Fallen vegetation adds visual complexity to train tracks' visual appearance recognition with respect to intelligent perception, which may impede any trained computer vision agent perception capability.		Unsafe_Scenario5.3 <code>{{fallen_vegetation_over_train_tracks_structure}}[follow]_{{passing_train_season_and_wind}},</code> <code>{{fallen_vegetation_over_train_tracks_structure}}[impede]_{{any_trained_computer_vision_agent_perception_capability}},</code> <code>{{fallen_vegetation_over_train_tracks_structure}}[obstruct]_{{train_tracks_structure_visual_appearance}}</code>
5.4.Vegetation around the fence adds visual complexity to the fence's visual recognizability, which may impede any trained computer vision agent perception capability.	Covered in 4.4	Covered in 4.4
5.5.Fallen vegetation, such as leaves and pollen, adds visual complexity to electrical powerlines' recognizability,		Unsafe_Scenario5.5 <code>{{fallen_vegetation_over_train_tracks_structure}}[follow]_{{passing_train_season_and_wind}},</code> <code>{{fallen_vegetation_over_train_tracks_structure}}[impede]_{{any_trained_computer_vision_agent_perception_capability}},</code> <code>{{fallen_vegetation_over_train_tracks_structure}}[obstruct]_{{powered_powerlines_cables_structure_visual_appearance}},</code>

which may impede any trained computer vision agent perception capability.		<code>{{fallen_vegetation_over_train_tracks_structure}}_generate_{{leaves_&_pollen}}</code>
6.3.Electric powerlines add visual complexity to train tracks' visual appearance, which may impede any trained computer vision agent perception capability.	Covered in 3.5	Covered in 3.5
6.4.Electric powerlines add visual complexity to train tracks fence visual appearance recognition, which may impede any trained computer vision agent perception capability.	Covered in 4.5	Covered in 4.5
6.5.Electric powerlines add visual complexity to vegetation around the fence, which may impede any trained computer	Covered in 5.5	Covered in 5.5

vision agent perception capability.		
7.1 Police Officers are incapable of catching adversarial drones.		Unsafe_Scenario7.1 $\{ \{ \text{police_force_existing_capability} \} _[-\text{monitor}] _ \{ \text{roaming_adversarial_drone} \} \},$ $\{ \{ \text{police_force_existing_capability} \} _ [+ \text{support}] _ \{ \text{safe_secure_train_track_zone} \} \},$ $\{ \{ \text{police_force_existing_capability} \} _ [+ \text{direct}] _ \{ \text{police_officer_situational_awareness} \} \}$
7.2 Police Officers avoid getting hit by train.		Unsafe_Scenario7.2 $\{ \{ \text{police_officer_situational_awareness} \} _ [+ \text{monitor}] _ \{ \text{train_motion_dynamics} \} \},$ $\{ \{ \text{police_officer_situational_awareness} \} _ [+ \text{support}] _ \{ \text{safe_secure_train_track_zone} \} \},$ $\{ \{ \text{police_officer_situational_awareness} \} _ [+ \text{support}] _ \{ \text{safe_secure_police_officer} \} \}$
8.1 Adversarial drone avoids capture by police.		Unsafe_Scenario8.1 $\{ \{ \text{roaming_adversarial_drone} \} _ [- \text{avoid}] _ \{ \text{police_officer_situational_awareness} \} \},$ $\{ \{ \text{roaming_adversarial_drone} \} _ [- \text{instigate}] _ \{ \text{birds_wasps_attack_expose_adversarial_drone} \} \},$ $\{ \{ \text{roaming_adversarial_drone} \} _ [- \text{avoid}] _ \{ \text{police_officer_capture_adversarial_drone} \} \}$
8.2 Adversarial drones fly over the train, making it hard for police to manage it.		Unsafe_Scenario8.2 $\{ \{ \text{roaming_adversarial_drone} \} _ [- \text{avoid}] _ \{ \text{police_officer_situational_awareness} \} \},$ $\{ \{ \text{roaming_adversarial_drone} \} _ [- \text{impede}] _ \{ \text{police_officer_capture_adversarial_drone} \} \},$ $\{ \{ \text{roaming_adversarial_drone} \} _ [- \text{fly_over}] _ \{ \text{free_space_above_train_roof} \} \}$

9.1 Train presence causes an obstacle to police officers' attempts to capture adversarial drones.		Unsafe_Scenario9.1 <pre> {stationary_train}_[support]_{roaming_adversarial_drone} , {stationary_train}_[impede]_{police_officer_capture_adversarial_drone} , {stationary_train}_[obstruct]_{police_officer_movements} </pre>
9.2 The train allows the adversarial drone to hover on top of it, making it harder for police officers to reach drones.		Unsafe_Scenario9.2 <pre> {roaming_adversarial_drone}_[appreciate]_{powered_powerlines_cables_structure_visual_appearance} , {roaming_adversarial_drone}_[-impede]_{police_officer_capture_adversarial_drone} , {roaming_adversarial_drone}_[-hover]_{free_space_above_train_roof} </pre>

Step 3.2) Predict the extended list of emergent AIC interactions scenarios

We then capture the interactions in the figure above using the following SECoT. Note we use the interaction format of

|{situation}_[action]_{situation}| to help us with automating extraction of factors. The key activity is we would do the AIC factorisation to each interaction in order to reveal the hidden interactions.

Table J.7 AIC Extended Scenarios

Step 1: Select observed outcome behaviour: Unsafe problematic activities	Unsafe_Scenario1.6 =The moving train complicates police officers' actions $\{ \{ \text{unsafe_moving_train} \} _ [\text{ignore}] _ \{ \text{unaware_train_driver_situational_awareness} \} \}$ $\{ \{ \text{unsafe_moving_train} \} _ [\text{-lead}] _ \{ \text{unsafe_passengers} \} \}$ $\{ \{ \text{unsafe_moving_train} \} _ [\text{-direct}] _ \{ \text{train_motion_dynamics} \} \}$ Selected interactions for further analysis : 1) $\{ \{ \text{unsafe_moving_train} \} _ [\text{-approach}] _ \{ \text{unsafe_train_tracks_zone_structure} \} \}$ 2) $\{ \{ \text{unsafe_moving_train} \} _ [\text{+approach}] _ \{ \text{roaming_adversarial_drone} \} \}$
step 2: observed system (obs)	Step 3: Inquired interaction
moving_train	1) $\{ \{ \text{unsafe_moving_train} \} _ [\text{-approach}] _ \{ \text{unsafe_train_tracks_zone_structure} \} \}$ Note that we used (-) because, currently, the moving train worsens the situation
step 5: auxiliary influence interaction	step 6: auxiliary control interaction
influence goal: 1. $\{ \{ \text{unsafe_moving_train} \} _ [\text{maintains}] _ \{ \text{safe_passengers_and_goods} \} \}$ influence actions: 1. $\{ \{ \text{unsafe_moving_train} \} _ [\text{inform}] _ \{ \text{train_driver_situational_awareness} \} \}$	control goals: 1. $\{ \{ \text{unsafe_moving_train} \} _ [\text{control}] _ \{ \text{onboard_train_safety_system} \} \}$ 2. $\{ \{ \text{unsafe_moving_train} \} _ [\text{control}] _ \{ \text{alert_systems} \} \}$ control actions: for {onboard_train_safety_systems} (preventing unsafe passengers)

2. {{unsafe_moving_train}}_prevent_{{unsafe_unsecured_passengers_and_goods}}	1. {{unsafe_moving_train}}_activate_{{onboard_train_safety_systems}} for {{alert_systems}} (informing the train driver): 2. {{unsafe_moving_train}}_trigger_{{alert_systems}} 3. {{unsafe_moving_train}}_alert_{{train_driver_situational_awareness}}
step 8: Predicted outcome emergence contribution	
factors or features that manifest influence interaction: ['moving_train', 'safe_passengers_and_goods', 'moving_train', 'safe_passengers_and_goods', 'moving_train', 'train_driver_situational_awareness', 'moving_train', 'unsecured_passengers_and_goods'] factors or features that manifest control interaction: ['moving_train', 'onboard_train_safety_systems', 'moving_train', 'alert_systems', 'onboard_train_safety_systems', 'moving_train', 'onboard_train_safety_systems', 'alert_systems'] factors or features that manifest appreciation interaction: ['moving_train', 'track_zone_monitoring_systems', 'moving_train', 'track_zone_monitoring_systems_data', 'moving_train', 'timely_track_zone_monitoring_systems_data', 'track_zone_monitoring_systems_data_gaps_or_inconsistencies']	
step 2: observed system (obs) moving_train, roaming_adversarial_drone	Step 3: Inquired interaction 2){{unsafe_moving_train}}_+approach_{{roaming_adversarial_drone}} Note that we used (-) because, currently, the moving train improves the train driver's situational awareness, which will achieve its goals.
step 5: auxiliary influence interaction	step 6: auxiliary control interaction

influence goal:

{unsafe_moving_train}_[influence]_{roaming_adversarial_drones}

influence actions:

1. communication (ground-based security systems):

1.1 {unsafe_moving_train}_[inform]_{available_ground_based_security_systems}

1.2

{unsafe_moving_train}_[communicate]_{roaming_adversarial_drones_location_data}

1.3

{unsafe_moving_train}_[communicate]_{roaming_adversarial_drones_information}

2. coordination (emergency response teams):

2.1

{unsafe_moving_train}_[trigger_communication]_{unaware_emergency_response_teams}

control goal 1:

1. {unsafe_moving_train}_[control]_{available_ground_based_security_systems}
2. {unsafe_moving_train}_[control]_{roaming_adversarial_drones_location_data}
3. {unsafe_moving_train}_[control]_{roaming_adversarial_drones_information}

control actions 1:**for communication interface:**

1. {unsafe_moving_train}_[establish_connection]_{available_ground_based_security_systems}
2. {unsafe_moving_train}_[transmit_data_packets]_{onboard_train_communication_interface}

for data processing unit:

3. {unsafe_moving_train}_[filter]_{roaming_adversarial_drones_size_data}
4. {unsafe_moving_train}_[filter]_{roaming_adversarial_drones_location_data}
5. {unsafe_moving_train}_[filter]_{roaming_adversarial_drones_flying_behavior}
6. {unsafe_moving_train}_[prioritize]_{roaming_adversarial_drones_criticality}
7. {unsafe_moving_train}_[format]_{roaming_adversarial_drones_information}

control goal 2:

4. {unsafe_moving_train}_[trigger_communication]_{unaware_emergency_response_teams}

control actions 2:**for communication interface:**

5. {unsafe_moving_train}_[establish_connection]_{designated_emergency_response_teams}
6. {unsafe_moving_train}_[transmit]_{onboard_train_communication_interface}

step 8: Predicted outcome emergence contrib

factors or features that manifest influence interaction:

['moving_train', 'roaming_adversarial_drones', 'moving_train', 'available_ground_based_security_systems', 'moving_train', 'roaming_adversarial_drones_location_data', 'moving_train', 'roaming_adversarial_drones_size_data', 'moving_train', 'roaming_adversarial_drones_location_data', 'moving_train', 'roaming_adversarial_drones_flight_data', 'moving_train', 'roaming_adversarial_drones_information_transmitted_data', 'moving_train', 'unaware_emergency_response_teams', 'moving_train', 'designated_emergency_response_teams']

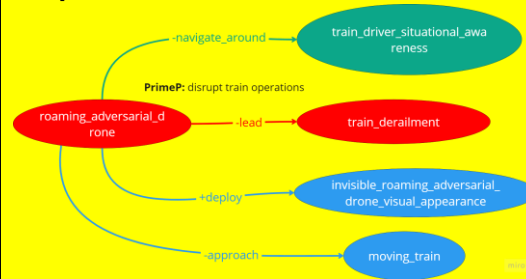
factors or features that manifest control interaction:

['moving_train', 'available_ground_based_security_systems', 'moving_train', 'roaming_adversarial_drones_location_data', 'moving_train', 'roaming_adversarial_drones_size_data', 'moving_train', 'roaming_adversarial_drones_location_data', 'moving_train', 'roaming_adversarial_drones_flight_data', 'moving_train', 'roaming_adversarial_drones_information_transmitted_data', 'moving_train', 'unaware_emergency_response_teams', 'moving_train', 'designated_emergency_response_teams']

factors or features that manifest appreciation interaction:

['moving_train', 'available_ground_based_security_systems', 'moving_train', 'reliable_railways_traffic_control', 'moving_train', 'accurate_track_zone_monitoring_systems', 'moving_train', 'alternative_communication_channels', 'moving_train', 'track_zone_monitoring_systems_data_plausibility', 'moving_train', 'ground_based_security_systems_receive_data']

Step 1: Select observed outcome behaviour: Unsafe problematic activities



{roaming_adversarial_drone}_[-approach]_{unsafe_moving_train}},

unsafe_scenario2.1

{roaming_adversarial_drone}_{+manage}_{invisible_roaming_adversarial_drone_visual_appearance}},

{roaming_adversarial_drone}_[-lead]_{train_derailment}},

{roaming_adversarial_drone}_[-navigate_around]_{train_driver_situational_awareness}},

Step 2: observed system (obs)

roaming_adversarial_drone

Step 3: Inquired interaction

{roaming_adversarial_drone}_[-approach]_{unsafe_moving_train}}

Step 5: auxiliary influence interaction

Step 6: auxiliary control interaction

influence goals: 1. {{roaming_adversarial_drone}}[influence]_{{train_driver_situational_awareness}} 2. {{roaming_adversarial_drone}}[influence]_{{onboard_train_safety_systems}} influence actions: 3. {{roaming_adversarial_drone}}[maintain]_{{low_altitude_roaming_adversarial_drone}} 4. {{roaming_adversarial_drone}}[employ]_{{roaming_adversarial_drone_camouflage}} 5. {{roaming_adversarial_drone}}[suppress]_{{roaming_adversarial_drone_noise_emissions}} 6. {{roaming_adversarial_drone}}[exploit]_{{moving_train_blind_spots}}	control goals: 1. {{roaming_adversarial_drone}}[control]_{{altitude_control_systems}} 2. {{roaming_adversarial_drone}}[control]_{{camouflage_mechanisms}} 3. {{roaming_adversarial_drone}}[control]_{{roaming_adversarial_drone}} 4. {{roaming_adversarial_drone}}[control]_{{roaming_adversarial_drone}} control actions: 1. {{roaming_adversarial_drone}}[adjust]_{{altitude_parameters}} 2. {{roaming_adversarial_drone}}[activate]_{{camouflage_mechanisms}} 3. {{roaming_adversarial_drone}}[reduce]_{{roaming_adversarial_drone}} 4. {{roaming_adversarial_drone}}[optimize]_{{path_planning_algorithms}} 5. {{roaming_adversarial_drone}}[monitor]_{{bad_weather}} 6. {{roaming_adversarial_drone}}[adapt]_{{adversarial_drone_responses}}
step 8: Predicted outcome emergence contribution	
factors or features that manifest influence interaction: ['roaming_adversarial_drone', 'train_driver_situational_awareness', 'roaming_adversarial_drone', 'onboard_train_safety_systems', 'roaming_adversarial_drone', 'low_altitude_roaming_adversarial_drone', 'roaming_adversarial_drone_noise_emissions', 'roaming_adversarial_drone', 'moving_train_blind_spots']	

factors or features that manifest control interaction:

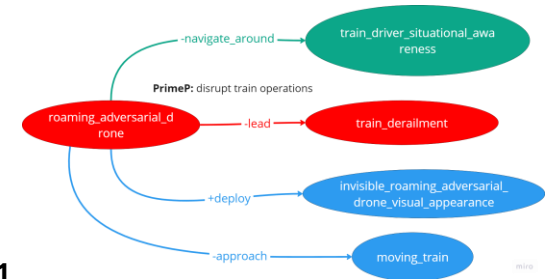
['roaming_adversarial_drone', 'altitude_control_systems', 'roaming_adversarial_drone', 'camouflage_mechanisms', 'roaming_adversarial_drone', 'roaming_adversarial_drone', 'roaming_adversarial_drone', 'altitude_parameters', 'roaming_adversarial_drone', 'camouflage_mechanisms', 'roaming_adversarial_drone', 'roaming_adversarial_drone', 'roaming_adversarial_drone', 'adversarial_drone_response_strategies']

factors or features that manifest appreciation interaction:

['roaming_adversarial_drone', 'weather_conditions', 'roaming_adversarial_drone', 'train_schedule_timing', 'roaming_adversarial_drone', 'train_tracks_maintenance', 'train_schedule_timing', 'roaming_adversarial_drone', 'train_tracks_maintenance_activities', 'roaming_adversarial_drone', 'track_zone_obstacles', 'roaming_adversarial_drone', 'adversarial_drone_dangerous_navigation_around_obstacles']

Step 1: Select observed outcome behaviour: Unsafe problematic activities

unsafe_scenario2.1

**Step 2: observed system (obs)**

roaming_adversarial_drone

Step 3: Inquired interaction

{roaming_adversarial_drone}_[+manage]_{invisible_roaming_adversarial_drone_visual_appearance}

	The roaming adversarial drone manages (controls or influences) its own visual signature, low observability, thereby evading detection by visual perception systems (e.g., radar, thermal, and monitoring systems).
Step 5: auxiliary influence interaction	Step 6: auxiliary control interaction
Influence Goal: 1. {{roaming_adversarial_drone}}[influence]_{{inadequate_eagle_drone_perception_reliability}} 2. {{roaming_adversarial_drone}}[influence]_{{inadequate_track_zone_monitoring_systems_detection_capability}} 3. {{roaming_adversarial_drone}}[influence]_{{unaware_train_driver_situational_awareness}} Influence Actions: 1. For {{inadequate_eagle_drone_perception_reliability}}: 1.1 {{roaming_adversarial_drone}}[reduce]_{{detectable_visual_signature}} 1.2 {{roaming_adversarial_drone}}[exploit]_{{low_contrast_environmental_conditions}} 1.3 {{roaming_adversarial_drone}}[mimic]_{{benign_environmental_object_appearance}} 2. For {{inadequate_track_zone_monitoring_systems_detection_capability}}: 2.1 {{roaming_adversarial_drone}}[blend]_{{ambient_background_visual_texture}} 2.2 {{roaming_adversarial_drone}}[obscure]_{{characteristic_drone_shape_features}} 3. For {{unaware_train_driver_situational_awareness}}: 3.1 {{roaming_adversarial_drone}}[delay]_{{timely_threat_notification}} 3.2 {{roaming_adversarial_drone}}[degrade]_{{reliable_security_alert_information}}	Control Goal 1: 1. {{roaming_adversarial_drone}}[control]_{{adaptive_camouflage}} 2. {{roaming_adversarial_drone}}[control]_{{low_observability_coating}} 3. {{roaming_adversarial_drone}}[control]_{{dynamic_surface_pattern_generator}} Control Actions 1: For {{adaptive_camouflage_system}}: 1. {{roaming_adversarial_drone}}[activate]_{{adaptive_camouflage}} 2. {{roaming_adversarial_drone}}[adjust]_{{real_time_camouflage}} 3. {{roaming_adversarial_drone}}[synchronize]_{{background_environment}} For {{low_observability_coating_material}}: 4. {{roaming_adversarial_drone}}[apply]_{{low_reflectivity_coating}} 5. {{roaming_adversarial_drone}}[minimize]_{{specular_reflection}} For {{dynamic_surface_pattern_generator}}: 6. {{roaming_adversarial_drone}}[generate]_{{adaptive_visual_pattern}} 7. {{roaming_adversarial_drone}}[modulate]_{{surface_color_distribution}} 8. {{roaming_adversarial_drone}}[match]_{{ambient_lighting_conditions}}

	Control Goal 2: 4. $\{\{roaming_adversarial_drone\}_{control}\}_{flight_trajectory_profile}$ 5. $\{\{roaming_adversarial_drone\}_{control}\}_{timing_intrusion_window}$ Control Actions 2: For $\{flight_trajectory_profile\}$: 9. $\{\{roaming_adversarial_drone\}_{select}\}_{low_visibility_flight_path}$ 10. $\{\{roaming_adversarial_drone\}_{avoid}\}_{high_illumination_zones}$ 11. $\{\{roaming_adversarial_drone\}_{exploit}\}_{visual_occlusion_opportunities}$ For $\{timing_intrusion_window\}$: 12. $\{\{roaming_adversarial_drone\}_{calculate}\}_{optimal_intrusion_time}$ 13. $\{\{roaming_adversarial_drone\}_{delay}\}_{intrusion_until_low_detection_probability}$ 14. $\{\{roaming_adversarial_drone\}_{execute}\}_{time_coordinated_perception_action}$
step 8: Predicted outcome emergence contribution	
factors or features that manifest influence interaction: $['roaming_adversarial_drone', 'inadequate_eagle_drone_perception_reliability', 'roaming_adversarial_drone', 'inadequate_track_zone_monitoring_systems_detection_probability', 'roaming_adversarial_drone', 'detectable_visual_signature', 'roaming_adversarial_drone', 'low_contrast_environmental_conditions', 'roaming_adversarial_drone', 'background_visual_texture', 'ambient_background_visual_texture', 'roaming_adversarial_drone', 'characteristic_drone_shape_features', 'unaware_train_driver_situational_awareness', 'roaming_adversarial_drone', 'unaware_train_driver_situational_awareness']$ factors or features that manifest control interaction:	

['roaming_adversarial_drone', 'adaptive_camouflage_system', 'roaming_adversarial_drone', 'low_observability_coating_material', 'roaming_adversarial_drone', 'dynamic_surface_pattern_generator', 'roaming_adversarial_drone', 'real_time_camouflage_pattern', 'roaming_adversarial_drone', 'background_environmental_visual_data', 'low_observability_coating_material', 'dynamic_surface_pattern_generator', 'roaming_adversarial_drone', 'adaptive_visual_pattern', 'roaming_adversarial_drone', 'surface_color_distribution', 'roaming_adversarial_drone', 'timing_intrusion_window', 'flight_trajectory_profile', 'roaming_adversarial_drone', 'low_visibility_flight_path', 'roaming_adversarial_drone', 'high_illumination_zones', 'roaming_adversarial_drone', 'intrusion_until_low_detection_probability', 'roaming_adversarial_drone', 'time_coordinated_penetration']

factors or features that manifest appreciation interaction:

['roaming_adversarial_drone', 'current_environmental_lighting_conditions', 'roaming_adversarial_drone', 'active_eagle_drone_patrol_pattern', 'roaming_adversarial_drone', 'ambient_illumination_intensity', 'roaming_adversarial_drone', 'shadow_zone_locations', 'roaming_adversarial_drone', 'low_contrast_background_textures', 'active_eagle_drone_sensor_coverage_gaps', 'roaming_adversarial_drone', 'predictable_patrol_schedule', 'available_visual_occlusion_features', 'roaming_adversarial_drone', 'fence_structure_occlusion_potential', 'roaming_adversarial_drone', 'adequate_concealment_coverage']

For simplicity, we will stop the AIC analysis here, and we focus on the high-level unsafe problematic interactions:

Unsafe_Scenario1.1 =

{unsafe_moving_train}[-approach]_{unsafe_train_tracks_zone_structure},
 {unsafe_moving_train}[-ignore]_{unaware_train_driver_situational_awareness},
 {unsafe_moving_train}[-lead]_{unsafe_unsecured_passengers_and_goods}

unsafe_scenario1.4 = {unsafe_moving_train}[-agitates]_{random_vegetation_motion},

{unsafe_moving_train}[-induces]_{traveling_train_induced_air_turbulence_and_vibration},

{unsafe_moving_train}[-pass_through]_{fence_to_train_open_space},

{unsafe_moving_train}[-impede]_{any_trained_computer_vision_agent_perception_capability},

```

unsafe_scenario1.5 = [{unsafe_moving_train}_[manage]_{train_electric_power_supply}],
[unsafe_moving_train]_[deploy]_{extended_train_pantograph}],
[unsafe_moving_train]_[ignore]_{free_space_above_train_roof}],
[{extended_train_pantograph}_[impede]_{any_trained_computer_vision_agent_perception_capability}],
[{extended_train_pantograph}_[trigger]_{powered_powerlines_cables_structure_sparks}],
[{extended_train_pantograph}_[chafe_&_wobble]_{powered_powerlines_cables_structure_visual_appearance}]

```

unsafe_scenario2.1

```

[roaming_adversarial_drone]_[+manage]_{invisible_roaming_adversarial_drone_visual_appearance}],
[roaming_adversarial_drone]_[lead]_{train_derailment}],
[roaming_adversarial_drone]_[navigate_around]_{train_driver_situational_awareness}],

```

unsafe_scenario2.2

```

[{roaming_adversarial_drone_perception_capability}_[trace]_{visible_train_tracks_appearance}],
[{roaming_adversarial_drone_perception_capability}_[lead]_{train_derailment}],
[{roaming_adversarial_drone_perception_capability}_[govern]_{roaming_adversarial_drone_flight_dynamics}]

```

unsafe_scenario2.3

```

(unsafe_scenario2.2

```

```
{roaming_adversarial_drone}_[-detect]_{visible_train_tracks_zone_fence_structure}},
```

```
{roaming_adversarial_drone}_[-lead]_{train_derailment}},
```

```
{roaming_adversarial_drone}_[+govern]_{roaming_adversarial_drone_flight_dynamics}},
```

```
{roaming_adversarial_drone}_[-approach]_{train_tracks_zone_fence_structure}}
```

.unsafe_scenario2.4

```
{roaming_adversarial_drone}_[+govern]_{roaming_adversarial_drone_flight_dynamics}},
```

```
{roaming_adversarial_drone}_[-recognise]_{visible_vegetation_visual_appearance}},
```

```
{roaming_adversarial_drone}_[-lead]_{train_derailment}},
```

```
{roaming_adversarial_drone}_[-avoid]_{visible_vegetation_structure}}
```

unsafe_scenario2.6

```
{roaming_adversarial_drone}_[-recognise]_{powered_powerlines_cables_structure_visual_appearance}},
```

```
{roaming_adversarial_drone}_[-lead]_{train_derailment}},
```

```
{roaming_adversarial_drone}_[-avoid]_{powered_powerlines_cables_structure_visual_appearance}},
```

```
{roaming_adversarial_drone}_[+govern]_{roaming_adversarial_drone_flight_dynamics}}
```

.unsafe_scenario2.7

```
{roaming_adversarial_drone}_[ignore]_{powered_powerlines_cables_emf}},
```

```
{roaming_adversarial_drone}_[-recognise]_{powered_powerlines_cables_structure_visual_appearance}},
```

```

|{roaming_adversarial_drone}_[-collide]_{powered_powerlines_cables_structure}},
|{roaming_adversarial_drone}_[-lead]_{train_derailment}},
|{roaming_adversarial_drone}_[+govern]_{roaming_adversarial_drone_flight_dynamics}}
.unsafe_scenario3.2
|{unsafe_train_tracks_zone_structure}_[ignore]_{visibility_obstructive_objects}},
|{unsafe_train_tracks_zone_structure}_[lead]_{adversarial_drone_collision_with_train}},
|{unsafe_train_tracks_zone_structure}_[guide]_{roaming_adversarial_drone_perception_capability}}
unsafe_scenario3.4
|{unsafe_train_tracks_zone_structure}_[ignore]_{train_tracks_structure_visual_appearance}},
|{unsafe_train_tracks_zone_structure}_[impede]_{any_trained_computer_vision_agent_perception_capability}},
|{unsafe_train_tracks_zone_structure}_[complicates]_{vegetation_complex_shapes_sizes_visual_appearance}},
|{unsafe_train_tracks_zone_structure}_[complicates]_{visible_train_tracks_zone_fence_structure_visual_appearance}}
.unsafe_scenario3.5
|{unsafe_train_tracks_zone_structure}_[ignore]_{train_tracks_structure_visual_appearance}},
|{unsafe_train_tracks_zone_structure}_[impede]_{any_trained_computer_vision_agent_perception_capability}},
|{unsafe_train_tracks_zone_structure}_[complicates]_{visible_powered_powerlines_cables_structure_visual_appearance}}
unsafe_scenario4.2

```

{train_tracks_zone_fence_structure}_[ignore]_{visible_train_tracks_zone_fence_structure_visual_appearance}},

{train_tracks_zone_fence_structure}_[guide]_{roaming_adversarial_drone_perception_capability}},

{train_tracks_zone_fence_structure}_[demarcate]_{train_tracks_structure_visual_appearance}}

unsafe_scenario4.4

{vegetation_complexity_growth}_[follow]_{train_tracks_zone_fence_structure}},

{vegetation_complexity_growth}_[impede]_{any_trained_computer_vision_agent_perception_capability}},

{vegetation_complexity_growth}_[generate]_{vegetation_complex_shapes_sizes_visual_appearance}}

unsafe_scenario4.5

{train_tracks_zone_fence_structure}_[ignore]_{visible_powered_powerlines_cables_structure_visual_appearance}},

{train_tracks_zone_fence_structure}_[impede]_{any_trained_computer_vision_agent_perception_capability}},

{train_tracks_zone_fence_structure}_[complicates]_{visible_powered_powerlines_cables_structure_visual_appearance}}

unsafe_scenario5.1

{vegetation_complexity_growth}_[adapt]_{train_track_zone_structure}},

{vegetation_complexity_growth}_[complicate]_{unaware_train_driver_situational_awareness}},

{vegetation_complexity_growth}_[generate]_{vegetation_complex_shapes_sizes_visual_appearance}}

unsafe_scenario5.3

{fallen_vegetation_over_train_tracks_structure}_[follow]_{passing_train_season_&_wind}},

{{fallen_vegetation_over_train_tracks_structure}_[impede]_{any_trained_computer_vision_agent_perception_capability}},

{{fallen_vegetation_over_train_tracks_structure}_[obstruct]_{train_tracks_structure_visual_appearance}}|covered in 4.4

unsafe_scenario5.5

{{fallen_vegetation_over_train_tracks_structure}_[follow]_{passing_train_season_&_wind}},

{{fallen_vegetation_over_train_tracks_structure}_[impede]_{any_trained_computer_vision_agent_perception_capability}},

{{fallen_vegetation_over_train_tracks_structure}_[obstruct]_{powered_powerlines_cables_structure_visual_appearance}},

{{fallen_vegetation_over_train_tracks_structure}_[generate]_{leaves_&_pollen}}|covered in 3.5covered in 4.5covered in 5.5

unsafe_scenario7.1

{{police_force_existing_capability}_[-monitor]_{roaming_adversarial_drone}},

{{police_force_existing_capability}_[+support]_{safe_secure_train_track_zone}},

{{police_force_existing_capability}_[+direct]_{police_officer_situational_awareness}}

unsafe_scenario7.2

{{police_officer_situational_awareness}_[+monitor]_{train_motion_dynamics}},

{{police_officer_situational_awareness}_[+support]_{safe_secure_train_track_zone}},

{{police_officer_situational_awareness}_[+support]_{safe_secure_police_officer}}

unsafe_scenario8.1

{{roaming_adversarial_drone}_[-avoid]_{police_officer_situational_awareness}},

{roaming_adversarial_drone}[-instigate]_{birds_wasps_attack_expose_adversarial_drone}},

{roaming_adversarial_drone}[-avoid]_{police_officer_capture_adversarial_drone}}

unsafe_scenario8.2

{roaming_adversarial_drone}[-avoid]_{police_officer_situational_awareness}},

{roaming_adversarial_drone}[-impede]_{police_officer_capture_adversarial_drone}},

{roaming_adversarial_drone}[-fly_over]_{free_space_above_train_roof}}

unsafe_scenario9.1

{stationary_train}[_support]_{roaming_adversarial_drone}},

{stationary_train}[_impede]_{police_officer_capture_adversarial_drone}},

{stationary_train}[_obstruct]_{police_officer_movements}}

unsafe_scenario9.2

{roaming_adversarial_drone}[_appreciate]_{powered_powerlines_cables_structure_visual_appearance}},

{roaming_adversarial_drone}[-impede]_{police_officer_capture_adversarial_drone}},

{roaming_adversarial_drone}[-hover]_{free_space_above_train_roof}}

below is the list of the remaining discovered factors

['moving_train', 'unsafe_train_tracks_zone_structure', 'moving_train', 'unaware_train_driver_situational_awareness', 'moving_train',
'unsafe_unsecured_passengers_and_goods', 'moving_train', 'random_vegetation_motion', 'moving_train',
'traveling_train_induced_air_turbulence_and_vibration', 'moving_train', 'fence_to_train_open_space', 'moving_train',

'any_trained_computer_vision_agent_perception_capability', 'moving_train', 'train_electric_power_supply', 'moving_train',
 'extended_train_pantograph', 'moving_train', 'free_space_above_train_roof', 'extended_train_pantograph',
 'any_trained_computer_vision_agent_perception_capability', 'extended_train_pantograph',
 'powered_powerlines_cables_structure_sparks', 'extended_train_pantograph',
 'powered_powerlines_cables_structure_visual_appearance', 'roaming_adversarial_drone',
 'invisible_roaming_adversarial_drone_visual_appearance', 'roaming_adversarial_drone', 'train_derailment', 'roaming_adversarial_drone',
 'train_driver_situational_awareness', 'roaming_adversarial_drone_perception_capability', 'visible_train_tracks_appearance',
 'roaming_adversarial_drone_perception_capability', 'train_derailment', 'roaming_adversarial_drone_perception_capability',
 'roaming_adversarial_drone_flight_dynamics', 'roaming_adversarial_drone', 'visible_train_tracks_zone_fence_structure',
 'roaming_adversarial_drone', 'train_derailment', 'roaming_adversarial_drone', 'roaming_adversarial_drone_flight_dynamics',
 'roaming_adversarial_drone', 'train_tracks_zone_fence_structure', 'roaming_adversarial_drone',
 'roaming_adversarial_drone_flight_dynamics', 'roaming_adversarial_drone', 'visible_vegetation_visual_appearance',
 'roaming_adversarial_drone', 'train_derailment', 'roaming_adversarial_drone', 'visible_vegetation_structure', 'roaming_adversarial_drone',
 'powered_powerlines_cables_structure_visual_appearance', 'roaming_adversarial_drone', 'train_derailment', 'roaming_adversarial_drone',
 'powered_powerlines_cables_structure_visual_appearance', 'roaming_adversarial_drone', 'roaming_adversarial_drone_flight_dynamics',
 'roaming_adversarial_drone', 'powered_powerlines_cables_emf', 'roaming_adversarial_drone',
 'powered_powerlines_cables_structure_visual_appearance', 'roaming_adversarial_drone', 'powered_powerlines_cables_structure',
 'roaming_adversarial_drone', 'train_derailment', 'roaming_adversarial_drone', 'roaming_adversarial_drone_flight_dynamics',
 'train_tracks_structure', 'visibility_obstructive_objects', 'train_tracks_structure', 'adversarial_drone_collision_with_train',
 'train_tracks_structure', 'roaming_adversarial_drone_perception_capability', 'train_tracks_structure',
 'train_tracks_structure_visual_appearance', 'train_tracks_structure', 'any_trained_computer_vision_agent_perception_capability',
 'train_tracks_structure', 'vegetation_complex_shapes_sizes_visual_appearance', 'train_tracks_structure',
 'visible_train_tracks_zone_fence_structure_visual_appearance', 'train_tracks_structure', 'train_tracks_structure_visual_appearance',
 'train_tracks_structure', 'any_trained_computer_vision_agent_perception_capability', 'train_tracks_structure',
 'visible_powered_powerlines_cables_structure_visual_appearance', 'train_tracks_zone_fence_structure',
 'visible_train_tracks_zone_fence_structure_visual_appearance', 'train_tracks_zone_fence_structure',
 'roaming_adversarial_drone_perception_capability', 'train_tracks_zone_fence_structure', 'train_tracks_structure_visual_appearance',
 'vegetation_complexity_growth', 'train_tracks_zone_fence_structure', 'vegetation_complexity_growth',
 'any_trained_computer_vision_agent_perception_capability', 'vegetation_complexity_growth',
 'vegetation_complex_shapes_sizes_visual_appearance', 'train_tracks_zone_fence_structure',
 'visible_powered_powerlines_cables_structure_visual_appearance', 'train_tracks_zone_fence_structure',
 'any_trained_computer_vision_agent_perception_capability', 'train_tracks_zone_fence_structure',
 'visible_powered_powerlines_cables_structure_visual_appearance', 'vegetation_complexity_growth', 'train_track_zone_structure',


```
'vegetation_complexity_growth', 'unaware_train_driver_situational_awareness', 'vegetation_complexity_growth',
'vegetation_complex_shapes_sizes_visual_appearance', 'fallen_vegetation_over_train_tracks_structure', 'passing_train_season_&_wind',
'fallen_vegetation_over_train_tracks_structure', 'any_trained_computer_vision_agent_perception_capability',
'fallen_vegetation_over_train_tracks_structure', 'train_tracks_structure_visual_appearance',
'fallen_vegetation_over_train_tracks_structure', 'passing_train_season_&_wind', 'fallen_vegetation_over_train_tracks_structure',
'any_trained_computer_vision_agent_perception_capability', 'fallen_vegetation_over_train_tracks_structure',
'powered_powerlines_cables_structure_visual_appearance', 'fallen_vegetation_over_train_tracks_structure', 'leaves_&_pollen',
'police_force', 'roaming_adversarial_drone', 'police_force', 'safe_secure_train_track_zone', 'police_force',
'police_officer_situational_awareness', 'police_officer_situational_awareness', 'train_motion_dynamics',
'police_officer_situational_awareness', 'safe_secure_train_track_zone', 'police_officer_situational_awareness',
'safe_secure_police_officer', 'roaming_adversarial_drone', 'police_officer_situational_awareness', 'roaming_adversarial_drone',
'birds_wasps_attack_expose_adversarial_drone', 'roaming_adversarial_drone', 'police_officer_capture_adversarial_drone',
'roaming_adversarial_drone', 'police_officer_situational_awareness', 'roaming_adversarial_drone',
'police_officer_capture_adversarial_drone', 'roaming_adversarial_drone', 'free_space_above_train_roof', 'stationary_train',
'roaming_adversarial_drone', 'stationary_train', 'police_officer_capture_adversarial_drone', 'stationary_train', 'police_officer_movements',
'roaming_adversarial_drone', 'powered_powerlines_cables_structure_visual_appearance', 'roaming_adversarial_drone',
'police_officer_capture_adversarial_drone', 'roaming_adversarial_drone', 'free_space_above_train_roof']
```

A useful python script to help with capturing situations from the interactions analysis can be used:

```
# The prupsoe of the extraction function is to extract factors from text
# choose the type of parenthesis where factors are captured in output factors analysis string. The parentheses type is a string

def extract_text_in_parentheses(input_factors_analysis, open_parentheses_type, close_parentheses_type):
    # List to store extracted phrases
    extracted_phrases = []

    # Temporary string to build phrases
```

```
temp_phrase = ""
inside_parentheses = False

# Iterate through each character in the text
for char in input_factors_analysis:
    if char == open_parentheses_type:
        inside_parentheses = True
        temp_phrase = ""
    elif char == close_parentheses_type and inside_parentheses:
        inside_parentheses = False
        # Add the phrase to the list
        extracted_phrases.append(temp_phrase)
    elif inside_parentheses:
        temp_phrase += char

return extracted_phrases
AIC_analysis = input("input AIC relationship analysis")

factors = extract_text_in_parentheses(AIC_analysis,"{","}")

print("\n\n")

print(factors)
```

```
print(f"""\n\n the number of factors is: {len(extract_text_in_parentheses(AIC_analysis,"{","}"))}""")
```

J.1.4 Predictive Thinking Pipeline 4: Predict and Evaluate Problem Domain Factors and Assumptions.

Step 4.1) Perform the most and least frequent factor evaluation

To perform this analysis, we needed to extract all the factors mentioned above and count their frequencies. The frequency of mention indicates the degree of emphasis we place on their influence on our peripheral perception of the problem. The following is Python code that we used to do so:

```
import csv
from collections import Counter

def extract_text_in_parentheses():
    """
    This code accepts text of any length and searches for specific characters to extract the text located between those characters. We employ the following format
    to define interactions:
    ({roaming_adversarial_drone}_[appreciate]_{powered_powerlines_structure}), where a factor is defined within curly braces {} and an action is enclosed within
    square brackets []. This script has been utilised to extract the factors and compile them into a list. Please execute this script in a local Python environment.
    Input the text in the fields of steps 2 to 7. The code will ignore any format that is not enclosed in { }.
    """
    # Prompt the user to enter a text
    input_text = input("enter your text: ")

    # List to store extracted phrases
    extracted_phrases = []
```

```
# Temporary string to build phrases
temp_phrase = ""
inside_parentheses = False

# Iterate through each character in the text
for char in input_text:
    if char == '{': # choose the opening character
        inside_parentheses = True
        temp_phrase = ""
    elif char == '}' and inside_parentheses: # choose the closing character
        inside_parentheses = False
        # Add the phrase to the list
        extracted_phrases.append(temp_phrase)
    elif inside_parentheses:
        temp_phrase += char

return extracted_phrases

predicted_factors = extract_text_in_parentheses()
print("-----")
print("-----")
print("-----")
print("-----")
```

```
print(predicted_factors)

def generate_histogram_and_csv(data_list):
    # Count the frequency of each element in the list
    frequency_count = Counter(data_list)

    # Calculate total occurrences for influence level calculation
    total_occurrences = sum(frequency_count.values())

    # Create a list of dictionaries for CSV writing
    csv_data = []
    for factor, freq in frequency_count.items():
        influence_level = freq / total_occurrences
        csv_data.append({
            "Predicted Factor": factor,
            "Frequency": freq,
            "Influence Level": influence_level
        })

    # Sort the list by frequency in descending order
    csv_data.sort(key=lambda x: x["Frequency"], reverse=True)

    # Write data to CSV file
    with open(r'C:\Users\hal1e20\python_ver\python_projects\System Science\predicted_factors.csv', 'w', newline='') as file:
```

```
writer = csv.DictWriter(file, fieldnames=["Predicted Factor", "Frequency", "Influence Level"])
writer.writeheader()
writer.writerows(csv_data)

return r"C:\Users\hal1e20\python_ver\python_projects\System Science\predicted_factors.csv"

# Example usage with a list of elements
#example_data = ["factor1", "factor2", "factor1", "factor3", "factor2", "factor1"]
#csv_file_path = generate_histogram_and_csv(example_data)
csv_file_path = generate_histogram_and_csv(predicted_factors)

print(csv_file_path) # output the location of the CSV file that computes the frequency of factors during analysis
```

Step 4.2) Define the problematic scenarios that may emerge from contributing situations

Table J.8a unsafe train tracks problem domain problematic scenarios

ID	predicted situation	possible problematic scenario
1	roaming_adversarial_drone	Unauthorized UAS approaches rail assets undetected, triggering line closure and emergency response delays.
2	moving_train	Train enters work zone with incomplete protection, creating near-miss with maintenance crew.
3	train_tracks_structure	Undetected fastener fatigue leads to gauge spread and speed restriction or derailment risk.
4	any_trained_computer_vision_agent_perception_capability	Computer vision misclassifies hazard under low contrast, delaying alarms and operational response.
5	train_tracks_zone_fence_structure	Fence breach unnoticed; corridor access enables proximity to rolling stock and equipment.
6	fallen_vegetation_over_train_tracks_structure	Random movements of downed branches impact ML perception reliability, obstruct rails; braking distance is insufficient, causing overrun into obstruction.
7	police_officer_situational_awareness	Officers lack real-time site picture; response routes conflict with active rail movements.
8	powered_powerlines_cables_structure_visual_appearance	Visual confusion near catenaries leads to sensor misreads; surveillance assets miss true hazard.
9	train_derailment	Wheel climb on defect magnifies into multi-car derailment near public area.
10	vegetation_complexity_growth	Dense foliage masks hazards; inspections miss encroachment threatening clearance and sightlines.
11	available_ground_based_security_systems	Sensor outage creates blind zone; intrusion undetected until operations impacted.
12	roaming_adversarial_drone_flight_dynamics	Unstable UAS drift toward track centerline during train pass, prompting emergency stop.
13	roaming_adversarial_drone_perception_capability	UAS sensing error causes unpredictable loiter near catenary; safety buffer violated.

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14	extended_train_pantograph	Pantograph entangles debris; arcing event forces unscheduled power isolation.
15	onboard_train_safety_systems	Safety system fault masked by nuisance alarms; late braking command during obstruction.
16	police_officer_capture_adversarial_drone	Intercept attempt distracts from perimeter safety, allowing secondary security gap.
17	train_driver_situational_awareness	Driver misses low-profile trackside hazard amid distraction; late braking.
18	train_tracks_structure_visual_appearance	Visual clutter hides cracked components; inspection cycle misses early warning signs.
19	unaware_train_driver_situational_awareness	Fatigue reduces vigilance; temporary speed restriction sign is overlooked.
20	adaptive_camouflage_system	Camouflaged device blends with track furniture; delayed detection during patrol.
21	alert_systems	Alarm flood desensitizes staff; critical alert dismissed as nuisance.
22	free_space_above_train_roof	Object on roof undetected; removal delayed due to access constraints and safety.
23	police_force	Resource dispatch mismatched; critical location unguarded during peak exposure.
24	roaming_adversarial_drones_location_data	Inaccurate geolocation misdirects responders; true hazard continues unobserved.
25	stationary_train	Standing train blocks sightlines and crossings, increasing trespass and collision risk.
26	unsafe_unsecured_passengers_and_goods	Unrestrained items shift during braking; passenger injury risk increases.
27	vegetation_complex_shapes_sizes_visual_appearance	Irregular foliage creates false positives; operators miss genuine threat indicators.
28	visible_powered_powerlines_cables_structure_visual_appearance	Overhead line geometry misguides sensors; hazard tracking degraded near masts.
29	weather_conditions	Sudden squall reduces adhesion and visibility; braking and detection performance decline.

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30	active_eagle_drone_patrol_pattern	Predictable patrol routes leave temporal coverage gaps exploited by incidents.
31	available_visual_occlusion_features	Poles and signage create blind corridors; monitoring fails to track moving hazards.
32	camouflage_mechanisms	Camouflage near track furniture delays recognition; response mobilization slowed.
33	current_environmental_lighting_conditions	Backlight silhouettes hide small objects; detection algorithms underperform.
34	dynamic_surface_pattern_generator	Changing patterns defeat simple detectors; surveillance confidence drops below alert threshold.
35	flight_trajectory_profile	Low-altitude path aligns with occluders; hazard proximity unnoticed until late.
36	inadequate_eagle_drone_perception_reliability	Patrol drone drops tracks intermittently; missed alerts during critical window.
37	inadequate_track_zone_monitoring_systems_detection_capability	Sensor range limits leave access points unobserved; delayed intrusion discovery.
38	invisible_roaming_adversarial_drone_visual_appearance	Low-visibility finish reduces human spotting; operations continue with unseen proximity.
39	low_observability_coating_material	Anti-glare surfaces lessen camera pickup; surveillance relies on less timely channels.
40	onboard_train_communication_interface	Misconfiguration enables message loss; safety alerts fail to reach crew.
41	passing_train_season_&_wind	Seasonal debris plus slipstream lofts objects onto track; strike hazard.
42	roaming_adversarial_drones_information	Incomplete intel on UAS intent causes conservative slowdowns and service disruption.
43	safe_passengers_and_goods	Overcrowding at platform edges increases incident likelihood during delays.
44	safe_secure_train_track_zone	Temporary works reduce barrier integrity; unauthorized access increases.
45	timing_intrusion_window	Incident coincides with shift change; response latency increases.

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46	track_zone_obstacles	Unreported obstacle within clearance envelope forces emergency braking.
47	train_schedule_timing	Timetable compression reduces recovery margins; cascading delays amplify risk.
48	train_tracks_maintenance_activities	Worksite miscommunication leaves equipment foul of track when train arrives.
49	unaware_emergency_response_teams	Responders lack updated site layout; access route conflicts with live track.
50	unsafe_train_tracks_zone_structure	Subgrade washout creates hidden void; track buckles under load.
51	visible_train_tracks_zone_fence_structure_visual_appearance	Fence pattern confuses basic detectors; intrusion remains unnoticed longer.
52	accurate_track_zone_monitoring_systems	Calibration drift reduces accuracy; false clears accepted during audit.
53	accurate_track_zone_monitoring_systems_data	Timestamp errors corrupt feeds; operators act on stale situational data.
54	adaptive_visual_pattern	Adaptive textures reduce surveillance saliency; late manual spotting.
55	adequate_concealment_coverage	Overestimated cover leaves hazard exposed to passengers before detection.
56	adversarial_drone_collision_with_train	Unexpected object strike damages equipment; investigation halts services.
57	adversarial_drone_dangerous_navigation_around_obstacles	Aggressive threading near structures results in crash debris on track.
58	adversarial_drone_response_strategies	Uncoordinated counter-UAS playbook triggers conflicting site actions.
59	birds_wasps_attack_expose_adversarial_drone	Network congestion compounds minor incident into systemwide delay and safety knock-ons.
60	alternative_communication_channels	Backup link untested; switchover fails during incident communications.
61	altitude_control_systems	Low-altitude control error places object inside clearance envelope near passing train.

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62	altitude_parameters	Unexpected altitude profile intersects overhead equipment; near contact event.
63	ambient_background_visual_texture	Background textures hide small hazards; automatic detection confidence drops.
64	ambient_illumination_intensity	High illuminance causes camera saturation; surveillance gaps appear.
65	ambient_lighting_conditions	Rapid dusk transition degrades recognition; alerts arrive too late.
66	background_environmental_visual_data	Seasonal texture shift breaks templates; detectors miss unusual intrusions.
67	benign_environmental_object_appearance	Everyday items misidentified as threats; real hazard filtered out.
68	characteristic_drone_shape_features	Altered silhouette evades shape-based filters; detection delayed.
69	communication_network_connectivity	Intermittent connectivity drops alarms; operators miss escalation cues.
70	designated_emergency_response_communication_system	Dedicated channel congests; site control receives fragmented instructions.
71	detectable_visual_signature	Suppressed signature delays recognition; hazard proximity increases unnoticed.
72	disruptive_adversarial_drones_threat_severity	Underestimated severity triggers insufficient response; operations continue unsafely.
73	eagle_drone_flight_trajectory	Patrol path misses siding; intrusion occurs in unvisited pocket.
74	eagle_drone_sensor_coverage_gaps	Known blind sectors remain unaddressed; repeated incursions exploit gap.
75	emergency_notification	Alert routing misconfigured; field staff not notified until after exposure.
76	fence_structure_occlusion_potential	Occlusion corridor hides movement; observers lose track near fence line.
77	fence_to_train_open_space	Gap between fence and train used as unseen corridor; late visual pickup.

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78	flight_path_based_on_weather	Gust-aligned path reduces detectability; hazard unobserved during wind peak.
79	ground_based_security_systems_received_data	Sensor fusion accepts corrupted feed; false situational picture guides actions.
80	high_illumination_zones	Bright zones create glare; CCTV misses small moving objects.
81	infrastructure_shadow_zones	Structure-cast shadows mask activity; perimeter patrol overlooks movement.
82	intrusion_until_low_detection_probability	Activity continues within modeled low-detection intervals; monitoring fails to trigger.
83	leaves_&_pollen	Airborne particulates confound optics; detectors suppress true positives.
84	low_altitude_roaming_adversarial_drone	Very low flight path remains below typical sightlines; late detection.
85	low_contrast_background_textures	Low-contrast backdrops degrade automated recognition; risk cues missed.
86	low_contrast_environmental_conditions	Dim scenes reduce VIO/detection reliability; hazard position uncertain.
87	low_reflectivity_coating_material	Low-reflectivity surfaces avoid glints; visual pickup degraded.
88	low_visibility_flight_path	Haze corridor hides movement; observers misjudge distance and presence.
89	moving_train_blind_spots	Car gap occlusion hides hazard until last moment; limited reaction time.
90	optimal_intrusion_timing	Activity synchronized with shift change escapes timely oversight.
91	path_planning_algorithms	Incorrect assumptions about patrol cycles route hazard through coverage gap.
92	police_officer_movements	Officer approach conflicts with live tracks; near-miss at crossing.
93	powered_powerlines_cables_emf	EMF noise degrades sensors; monitoring misses nearby hazard.

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94	powered_powerlines_cables_structure	Catenary alignments obscure sightlines; hazard tracked too late.
95	powered_powerlines_cables_structure_sparks	Arcing distracts operators; real hazard overlooked in same scene.
96	predictable_patrol_schedule	Routine timing enables repeated incidents between patrol passes.
97	random_vegetation_motion	Wind-driven motion triggers false alarms; true alerts lost in noise.
98	real_time_camouflage_pattern	Pattern updates disrupt classifier confidence; alarm threshold not reached.
99	reliable_railways_traffic_control	Localized system fault dispatches trains onto congested segment; margins shrink.
100	reliable_security_alert_information	Inaccurate metadata reduces alert credibility; delayed on-scene action.
101	roaming_adversarial_drone_camouflage	Environment-matched skins defeat quick visual checks; hazard proximity increases.
102	roaming_adversarial_drone_navigation_systems	GNSS-denied navigation drift causes unpredictable loiter near live tracks.
103	roaming_adversarial_drone_noise_emissions	Elevated ambient noise masks acoustic cues; observers miss approach.
104	roaming_adversarial_drone_noise_output	Sudden tonal peaks cause misplaced focus; true threat not located.
105	roaming_adversarial_drone_noise_suppression_systems	Quiet operation delays human detection; incident persists longer.
106	roaming_adversarial_drones	Multiple UAS reports saturate response; triage delays addressing actual hazard.
107	roaming_adversarial_drones_critical_data	Missing identifiers hinder correlation; the same hazard repeatedly “rediscovered.”
108	roaming_adversarial_drones_flying_behaviour_data	Erratic behavior flags overload analysts; prioritized response misassigned.
109	roaming_adversarial_drones_information_transmitted_data	Link encryption prevents rapid intel sharing; responders lack actionable location.

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110	roaming_adversarial_drones_size_data	Very small form factor evades sensors; close approach unnoticed.
111	routes_around_maintenance_activities	Temporary closures shift patrol patterns; unobserved corridor emerges.
112	safe_secure_police_officer	Poor cordon design exposes officers to train movement near incident.
113	shadow_zone_locations	Dynamic shadow bands conceal movement; surveillance misses brief exposures.
114	specular_reflection_intensity	Glints blind cameras intermittently; hazard track consistency lost.
115	surface_color_distribution	Color matching with ballast and sleepers reduces visibility; late manual detection.
116	time_coordinated_penetration	Activity synchronized with train pass masks noise and motion cues.
117	timely_threat_notification	Notification delays exceed response targets; hazard persists on scene.
118	timely_track_zone_monitoring_systems_data	Latent data feed misrepresents current state; wrong area patrolled.
119	track_zone_monitoring_systems	Misplacement leaves critical siding uncovered; incidents go unseen.
120	track_zone_monitoring_systems_data	Data quality issues drive false assurance; hazard proximity unrecognized.
121	track_zone_monitoring_systems_data_gaps_or_inconsistencies	Missing data intervals hide approach; alarms resume post-exposure.
122	track_zone_monitoring_systems_data_plausibility	Unchallenged anomalous values accepted as normal; operators overlook incident.
123	trackside_vegetation_density	High density blocks line-of-sight; inspection and patrol effectiveness reduced.
124	train_electric_power_supply	Power anomaly misinterpreted; unscheduled isolation disrupts operations and safety systems.
125	train_motion_dynamics	Slipstream effects loft debris into track; driver receives late cue.

126	train_track_zone_structure	Complex geometry creates unobserved corridors; monitoring coverage holes persist.
127	traveling_train_induced_air_turbulence_and_vibration	Vibration degrades cameras; short surveillance gaps coincide with hazards.
128	visibility_obstructive_objects	Stacked materials near right-of-way block cameras; alarms suppressed.
129	visible_train_tracks_appearance	Ballast texture confuses detectors; small objects go unnoticed.
130	visible_train_tracks_zone_fence_structure	Fence texture alignment hides motion; tracking drops intermittently.
131	visible_vegetation_structure	Branch lattice creates occlusion tunnel; hazard unseen until close.
132	visible_vegetation_visual_appearance	Seasonal foliage colors reduce contrast; human spotters miss cues.
133	visual_occlusion_opportunities	Systematic use of occluders breaks persistent tracking; response arrives late.

Define the assumptions made about factors

In this step, we used the Python script from step 4.1 and changed the format of interactions from: | interaction | to < interaction >. Then instructed the code to capture any text in between <>. In this version of the process, we include only the full interaction and prompt the architect to describe the associated assumption with the interaction.

Table J.9b unsafe train tracks problem domain assumptions

Assumption label	Predicted interaction	Architect Prediction (assumption)
1	{roaming_adversarial_drone}_[-lead]_ {train_derailment}	An assumption that roaming adversarial drones could potentially lead to train derailment.
2	{unsafe_moving_train}_[approach]_ {unsafe_train_tracks_zone_structure}	It is assumed that as a moving train approaches, it may inadvertently interact with unsafe or poorly structured train tracks zones.

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3	{unsafe_moving_train}_[prevent] _{unsafe_unsecured_passengers_and_goods}	The assumption that moving trains may be unintentionally prevent risks associated with unsafe or unsecured passengers and goods.
4	{roaming_adversarial_drone}_[+govern] _{roaming_adversarial_drone_flight_dynamics}	It is assumed that roaming adversarial drones can positively influence or govern their flight dynamics for certain operations.
5	{unsafe_moving_train}_[maintains] _{safe_passengers_and_goods}	Assuming that moving trains inherently maintain safety for passengers and goods during their operation.
6	{unsafe_moving_train}_[inform] _{train_driver_situational_awareness}	An assumption that moving trains provide information that enhances train driver situational awareness.
7	{unsafe_moving_train}_[communicate] _{roaming_adversarial_drones_location_data}	The assumption that moving trains communicate or exchange data related to the location of roaming adversarial drones.
8	{unsafe_moving_train}_[communicate] _{roaming_adversarial_drones_information}	It is assumed that moving trains communicate information related to roaming adversarial drones.
9	{unsafe_moving_train}_[trigger_communication] _{unaware_emergency_response_teams}	The assumption that moving trains trigger communication with emergency response teams who may be unaware of certain circumstances.
10	{roaming_adversarial_drone}_[maintain] _{low_altitude_roaming_adversarial_drone}	An assumption that invisible roaming adversarial drones maintain a low altitude during their operations.
11	{roaming_adversarial_drone}_[employ] _{roaming_adversarial_drone_camouflage}	The assumption that invisible roaming adversarial drones employ camouflage techniques during operations.
12	{roaming_adversarial_drone}_[suppress] _{roaming_adversarial_drone_noise_emissions}	Assuming that invisible roaming adversarial drones suppress noise emissions to avoid detection.

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13	{roaming_adversarial_drone}_[exploit] _{moving_train_blind_spots}	An assumption that invisible roaming adversarial drones exploit the blind spots of moving trains for their operations.
14	{roaming_adversarial_drone}_[monitor] _{bad_weather}	It is assumed that invisible roaming adversarial drones monitor weather conditions for operational advantages.
15	{roaming_adversarial_drone}_[-recognise] _{powered_powerlines_cables_structure_visual_appearance}	The assumption that roaming adversarial drones recognize powered powerlines cables based on their visual appearance.
16	{unsafe_train_tracks_zone_structure}_[ignore] _{train_tracks_structure_visual_appearance}	Assuming that train tracks structures ignore certain visual appearances during their operations.
17	{unsafe_train_tracks_zone_structure}_[impede] _{any_trained_computer_vision_agent_perception_capability}	An assumption that train tracks structures impede the perception capability of any trained computer vision agents.
18	{vegetation_complexity_growth}_[generate] _{vegetation_complex_shapes_sizes_visual_appearance}	It is assumed that vegetation complexity growth generates various shapes and sizes with distinct visual appearances.
19	{fallen_vegetation_over_train_tracks_structure}_[follow] _{passing_train_season_&_wind}	An assumption that fallen vegetation over train tracks structures follows the patterns influenced by passing train seasons and wind conditions.
20	{fallen_vegetation_over_train_tracks_structure}_[impede] _{any_trained_computer_vision_agent_perception_capability}	An assumption that fallen vegetation over train tracks impedes the perception capability of any trained computer vision agents.
21	{roaming_adversarial_drone}_[-avoid] _{police_officer_situational_awareness}	Assuming that roaming adversarial drones actively avoid detection by police officers to maintain situational awareness.
22	{roaming_adversarial_drone}_[-impede] _{police_officer_capture_adversarial_drone}	It is assumed that roaming adversarial drones impede efforts by police officers to capture them.

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23	{unsafe_moving_train}_[direct] _{unaware_train_driver_situational_awareness}	An assumption that moving trains directly impact the situational awareness of unaware train drivers.
24	{unsafe_moving_train}_[ignore] _{roaming_adversarial_drone}	Assuming that moving trains ignore the presence of invisible roaming adversarial drones during their operations.
25	{unsafe_moving_train}_[lead] _{unsafe_unsecured_passengers_and_goods}	The assumption that moving trains lead to unsafe conditions for unsecured passengers and goods.
26	{unsafe_moving_train}_[control] _{onboard_train_safety_systems}	It is assumed that moving trains control onboard safety systems to ensure passenger and crew safety.
27	{unsafe_moving_train}_[control] _{alert_systems}	Assuming that moving trains control alert systems to notify personnel of potential hazards or malfunctions.
28	{unsafe_moving_train}_[activate] _{onboard_train_safety_systems}	An assumption that moving trains activate onboard safety systems to prevent accidents and ensure operational safety.
29	{unsafe_moving_train}_[trigger] _{alert_systems}	Assuming that moving trains trigger alert systems to alert personnel of critical situations or emergencies.
30	{unsafe_moving_train}_[alert] _{train_driver_situational_awareness}	An assumption that moving trains alert train drivers to enhance their situational awareness during operations.
31	{unsafe_moving_train}_[appreciate] _{track_zone_monitoring_systems}	It is assumed that moving trains appreciate the presence and functionality of track zone monitoring systems.
32	{unsafe_moving_train}_[initiate_collection] _{track_zone_monitoring_systems_data}	An assumption that moving trains initiate the collection of data from track zone monitoring systems for operational insights.
33	{unsafe_moving_train}_[acknowledge] _{timely_track_zone_monitoring_systems_data}	Assuming that moving trains acknowledge the importance of timely data from track zone monitoring systems for operational decisions.

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34	{unsafe_moving_train}_[acknowledge] _{accurate_track_zone_monitoring_systems_data}	It is assumed that moving trains acknowledge the significance of accurate data from track zone monitoring systems for reliable operational insights.
35	{unsafe_moving_train}_[highlight] _{track_zone_monitoring_systems_data_gaps_or_inconsistencies }	An assumption that moving trains highlight gaps or inconsistencies in data collected from track zone monitoring systems for corrective actions.
36	{unsafe_moving_train}_[approach] _{roaming_adversarial_drone}	Assuming that moving trains unintentionally approach roaming adversarial drones during their operations.
37	{unsafe_moving_train}_[minimize] _{roaming_adversarial_drones}	An assumption that moving trains minimize the impact of roaming adversarial drones on their operations.
38	{unsafe_moving_train}_[inform] _{available_ground_based_security_systems}	It is assumed that moving trains inform personnel about the availability of ground-based security systems for enhanced safety measures.
39	{unsafe_moving_train}_[communicate] _{available_ground_based_security_systems}	Assuming that moving trains communicate with personnel about the availability and functionality of ground-based security systems.
40	{unsafe_moving_train}_[establish_connection] _{available_ground_based_security_systems}	An assumption that moving trains establish connections with available ground-based security systems for enhanced safety measures.
41	{unsafe_moving_train}_[transmit_data_packets] _{onboard_train_communication_interface}	Assuming that moving trains transmit data packets through the onboard train communication interface for operational communication.
42	{unsafe_moving_train}_[filter] _{roaming_adversarial_drones_size_data}	It is assumed that moving trains filter data related to the size of roaming adversarial drones for operational insights.

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43	{unsafe_moving_train}_[filter] _{roaming_adversarial_drones_location_data}	An assumption that moving trains filter data related to the location of roaming adversarial drones for operational insights.
44	{unsafe_moving_train}_[filter] _{roaming_adversarial_drones_flying_behaviour_data}	Assuming that moving trains filter data related to the flying behaviour of roaming adversarial drones for operational insights.
45	{unsafe_moving_train}_[prioritize] _{roaming_adversarial_drones_critical_data}	It is assumed that moving trains prioritize critical data related to roaming adversarial drones for immediate action.
46	{unsafe_moving_train}_[format] _{roaming_adversarial_drones_information_transmitted_data}	An assumption that moving trains format information related to roaming adversarial drones before transmitting it for clarity and accuracy.
47	{unsafe_moving_train}_[establish_connection] _{designated_emergency_response_communication_system}	Assuming that moving trains establish connections with designated emergency response communication systems for coordinated actions during emergencies.
48	{unsafe_moving_train}_[transmit] _{onboard_train_communication_interface} {emergency_notification}	An assumption that moving trains transmit emergency notifications through the onboard train communication interface for immediate response.
49	{unsafe_moving_train}_[appreciate] _{available_ground_based_security_systems}	It is assumed that moving trains appreciate the presence and functionality of available ground-based security systems for enhanced operational safety.
50	{unsafe_moving_train}_[appreciate] _{reliable_railways_traffic_control}	An assumption that moving trains appreciate the reliability of railways communication networks for operational efficiency and safety.
51	{unsafe_moving_train}_[appreciate] _{accurate_track_zone_monitoring_systems}	Assuming that moving trains appreciate the accuracy of track zone monitoring systems for enhanced safety and operational awareness.

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52	{unsafe_moving_train}_{monitor} _{available_ground_based_security_systems}	It is assumed that moving trains monitor available ground-based security systems for potential threats and safety measures.
53	{unsafe_moving_train}_{monitor} _{communication_network_connectivity}	An assumption that moving trains monitor communication network connectivity for reliable operational communication.
54	{unsafe_moving_train}_{establish} _{alternative_communication_channels}	Assuming that moving trains establish alternative communication channels for redundant operational communication.
55	{unsafe_moving_train}_{verify} _{track_zone_monitoring_systems_data_plausibility}	It is assumed that moving trains verify the plausibility of data from track zone monitoring systems for accuracy in decision-making.
56	{unsafe_moving_train}_{confirms} _{ground_based_security_systems_received_data}	An assumption that moving trains confirm the reception of data from ground-based security systems for timely response to threats.
57	{unsafe_moving_train}_{assess} _{disruptive_adversarial_drones_threat_severity}	Assuming that moving trains assess the severity of threats posed by disruptive adversarial drones for appropriate response measures.
58	{roaming_adversarial_drone}_{approach} _{unsafe_moving_train}	An assumption that invisible roaming adversarial drones approach moving trains for potential interference or observation goals.
59	{roaming_adversarial_drone}_{influence} _{train_driver_situational_awareness}	It is assumed that invisible roaming adversarial drones influence train driver situational awareness through their presence or actions.
60	{roaming_adversarial_drone}_{influence} _{onboard_train_safety_systems}	An assumption that invisible roaming adversarial drones influence onboard train safety systems for potential disruption or compromise.
61	{roaming_adversarial_drone}_{control} _{altitude_control_systems}	Assuming that invisible roaming adversarial drones can control altitude control systems to manipulate flight height.

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62	{roaming_adversarial_drone}_{control} _{camouflage_mechanisms}	An assumption that invisible roaming adversarial drones control camouflage mechanisms to avoid detection.
63	{roaming_adversarial_drone}_{control} _{roaming_adversarial_drone_noise_suppression_systems}	Assuming that invisible roaming adversarial drones control noise suppression systems to reduce their detectability.
64	{roaming_adversarial_drone}_{control} _{roaming_adversarial_drone_navigation_systems}	An assumption that invisible roaming adversarial drones control navigation systems to maneuver effectively.
65	{roaming_adversarial_drone}_{adjust} _{altitude_parameters}	Assuming that invisible roaming adversarial drones adjust altitude parameters to maintain covert operations.
66	{roaming_adversarial_drone}_{activate} _{camouflage_mechanisms}	An assumption that invisible roaming adversarial drones activate camouflage mechanisms to blend into the environment.
67	{roaming_adversarial_drone}_{reduce} _{roaming_adversarial_drone_noise_output}	Assuming that invisible roaming adversarial drones reduce noise output to avoid detection or suspicion.
68	{roaming_adversarial_drone}_{optimize} _{path_planning_algorithms}	An assumption that invisible roaming adversarial drones optimize path planning algorithms to evade detection or navigate efficiently.
69	{roaming_adversarial_drone}_{adapt} _{adversarial_drone_response_strategies}	Assuming that invisible roaming adversarial drones adapt their strategies to counter response measures.
70	{roaming_adversarial_drone}_{appreciate} _{bad_weather}	An assumption that invisible roaming adversarial drones appreciate weather conditions to adjust flight actions accordingly.
71	{roaming_adversarial_drone}_{appreciate} _{train_schedule_timing}	Assuming that invisible roaming adversarial drones appreciate train schedule timing to synchronize activities or avoid detection.
72	{roaming_adversarial_drone}_{appreciate} _{train_tracks_maintenance_activities}	An assumption that invisible roaming adversarial drones appreciate train tracks

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		maintenance activities to exploit vulnerabilities or avoid detection.
73	{roaming_adversarial_drone}_[appreciate] _{track_zone_obstacles}	Assuming that invisible roaming adversarial drones appreciate track zone obstacles to navigate or avoid detection.
74	{roaming_adversarial_drone}_[synchronize] _{train_schedule_timing}	An assumption that invisible roaming adversarial drones synchronize with train schedule timing to perform actions or avoid detection during critical periods.
75	{roaming_adversarial_drone}_[scan] _{train_tracks_maintenance_activities}	Assuming that invisible roaming adversarial drones scan train tracks maintenance activities to gather intelligence or plan actions.
76	{roaming_adversarial_drone}_[detect] _{track_zone_obstacles}	An assumption that invisible roaming adversarial drones detect track zone obstacles to adjust flight path or avoid collisions.
77	{roaming_adversarial_drone}_[adapt] _{flight_path_based_on_weather}	Assuming that invisible roaming adversarial drones adapt flight paths based on current weather conditions to maintain operational effectiveness.
78	{roaming_adversarial_drone}_[plan] _{routes_around_maintenance_activities}	An assumption that invisible roaming adversarial drones plan routes around train maintenance activities to avoid detection or disruption.
79	{roaming_adversarial_drone}_[navigate] _{adversarial_drone_dangerous_navigation_around_obstacles}	Assuming that invisible roaming adversarial drones navigate dangerous paths around obstacles to maintain covert operations.
80	{unsafe_moving_train}_[agitates] _{random_vegetation_motion}	An assumption that a moving train agitates random vegetation, potentially causing movement or disturbance in its vicinity.
81	{unsafe_moving_train}_[induces] _{traveling_train_induced_air_turbulence_and_vibration}	Assuming that a moving train induces air turbulence and vibration as it passes through, affecting surrounding environment.

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82	{unsafe_moving_train}_[pass_through] _{fence_to_train_open_space}	An assumption that a moving train can pass through openings in fences or barriers intended to contain its path.
83	{unsafe_moving_train}_[impede] _{any_trained_computer_vision_agent_perception_capability}	Assuming that a moving train might impede the perception capability of any trained computer vision agent in its path.
84	{unsafe_moving_train}_[manage] _{train_electric_power_supply}	An assumption that a moving train manages its electric power supply efficiently during operation.
85	{unsafe_moving_train}_[deploy] _{extended_train_pantograph}	Assuming that a moving train deploys an extended pantograph when required to maintain electrical contact with overhead wires.
86	{unsafe_moving_train}_[ignore] _{free_space_above_train_roof}	An assumption that a moving train ignores the presence of free space above its roof during its journey.
87	{extended_train_pantograph}_[impede] _{any_trained_computer_vision_agent_perception_capability}	Assuming that an extended train pantograph might impede the perception capability of any trained computer vision agent near it.
88	{extended_train_pantograph}_[trigger] _{powered_powerlines_cables_structure_sparks}	An assumption that an extended train pantograph might trigger sparks when in contact with powered power lines or cables.
89	{extended_train_pantograph}_[chafe_&_wobble] _{powered_powerlines_cables_structure_visual_appearance}	Assuming that an extended train pantograph might chafe and wobble when in contact with powered power lines or cables, affecting their visual appearance.
90	{roaming_adversarial_drone}_[+manage] _{invisible_roaming_adversarial_drone_visual_appearance}	An assumption that a roaming adversarial drone can manage its visual appearance to remain invisible to detection systems.
91	{roaming_adversarial_drone}_[navigate_around] _{train_driver_situational_awareness}	Assuming that a roaming adversarial drone navigates around obstacles, potentially obstructing train drivers' situational awareness.

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92	{roaming_adversarial_drone_perception_capability}_[-trace] _{visible_train_tracks_appearance}	An assumption that the perception capability of a roaming adversarial drone can trace visible train tracks' appearance.
93	{roaming_adversarial_drone_perception_capability}_[-lead] _{train_derailment}	Assuming that the perception capability of a roaming adversarial drone can lead to train derailment by misguiding or interfering with tracks.
94	{roaming_adversarial_drone_perception_capability}_[+govern] _{roaming_adversarial_drone_flight_dynamics}	An assumption that the perception capability of a roaming adversarial drone can govern its flight dynamics, ensuring stable and controlled movement.
95	{roaming_adversarial_drone}_[-detect] _{visible_train_tracks_zone_fence_structure}	Assuming that a roaming adversarial drone can detect visible structures like fences around train tracks, potentially hindering its movement.
96	{roaming_adversarial_drone}_[-approach] _{train_tracks_zone_fence_structure}	An assumption that a roaming adversarial drone approaches fence structures in train track zones, potentially leading to trespassing or obstruction.
97	{roaming_adversarial_drone}_[-recognise] _{visible_vegetation_visual_appearance}	Assuming that a roaming adversarial drone can recognize visible vegetation by its appearance, potentially altering its flight path.
98	{roaming_adversarial_drone}_[-avoid] _{visible_vegetation_structure}	An assumption that a roaming adversarial drone avoids visible vegetation structures, potentially altering its flight path to evade detection or collision.
99	{roaming_adversarial_drone}_[-avoid] _{powered_powerlines_cables_structure_visual_appearance}	Assuming that a roaming adversarial drone avoids powered power lines and cables based on their visual appearance, minimizing collision risks.
100	{roaming_adversarial_drone}_[ignore] _{powered_powerlines_cables_EMF}	An assumption that a roaming adversarial drone can manage its visual appearance to remain invisible to detection systems.

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101	{roaming_adversarial_drone}[_collide] _{powered_powerlines_cables_structure}	Assuming that a roaming adversarial drone navigates around obstacles, potentially obstructing train drivers' situational awareness.
102	{unsafe_train_tracks_zone_structure}[_ignore] _{visibility_obstructive_objects}	An assumption that the perception capability of a roaming adversarial drone can trace visible train tracks' appearance.
103	{unsafe_train_tracks_zone_structure}[_lead] _{adversarial_drone_collision_with_train}	Assuming that the perception capability of a roaming adversarial drone can lead to train derailment by misguiding or interfering with tracks.
104	{unsafe_train_tracks_zone_structure}[_guide] _{roaming_adversarial_drone_perception_capability}	An assumption that the perception capability of a roaming adversarial drone can govern its flight dynamics, ensuring stable and controlled movement.
105	{unsafe_train_tracks_zone_structure}[_complicates] _{vegetation_complex_shapes_sizes_visual_appearance}	Assuming that a roaming adversarial drone can detect visible structures like fences around train tracks, potentially hindering its movement.
106	{unsafe_train_tracks_zone_structure}[_complicates] _{visible_train_tracks_zone_fence_structure_visual_appearance}	An assumption that a roaming adversarial drone approaches fence structures in train track zones, potentially leading to trespassing or obstruction.
107	{unsafe_train_tracks_zone_structure}[_complicates] _{visible_powered_powerlines_cables_structure_visual_appearance}	Assuming that a roaming adversarial drone can recognize visible vegetation by its appearance, potentially altering its flight path.
108	{train_tracks_zone_fence_structure}[_ignore] _{visible_train_tracks_zone_fence_structure_visual_appearance}	An assumption that a roaming adversarial drone avoids visible vegetation structures, potentially altering its flight path to evade detection or collision.
109	{train_tracks_zone_fence_structure}[_guide] _{roaming_adversarial_drone_perception_capability}	Assuming that a roaming adversarial drone avoids powered power lines and cables based on their visual appearance, minimizing collision risks.

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110	{train_tracks_zone_fence_structure}_[demarcate] _{train_tracks_structure_visual_appearance}	An assumption that the train tracks zone fence structure demarcates the visual appearance of the train tracks' structure, potentially influencing how they are perceived visually.
111	{vegetation_complexity_growth}_[follow] _{train_tracks_zone_fence_structure}	Assuming that vegetation complexity growth follows the structure of train tracks zone fences, potentially affecting their maintenance and appearance.
112	{vegetation_complexity_growth}_[impede] _{any_trained_computer_vision_agent_perception_capability}	An assumption that vegetation complexity growth may impede the perception capability of any trained computer vision agents, affecting their ability to detect objects.
113	{train_tracks_zone_fence_structure}_[ignore] _{visible_powered_powerlines_cables_structure_visual_appearance}	Assuming that the train tracks zone fence structure ignores the visible appearance of powered power lines and cables, potentially affecting operational safety.
114	{train_tracks_zone_fence_structure}_[impede] _{any_trained_computer_vision_agent_perception_capability}	An assumption that the train tracks zone fence structure may impede the perception capability of any trained computer vision agents, affecting their accuracy in detecting objects.
115	{train_tracks_zone_fence_structure}_[complicates] _{visible_powered_powerlines_cables_structure_visual_appearance}	Assuming that the train tracks zone fence structure complicates the visible appearance of powered power lines and cables, potentially affecting their maintenance and safety.
116	{vegetation_complexity_growth}_[adapt] _{train_track_zone_structure}	An assumption that vegetation complexity growth adapts to the structure of train track zones, potentially influencing operational efficiency and maintenance practices.
117	{vegetation_complexity_growth}_[complicate] _{unaware_train_driver_situational_awareness}	Assuming that vegetation complexity growth complicates unaware train drivers' situational awareness, potentially impacting operational safety.

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118	{fallen_vegetation_over_train_tracks_structure}_[obstruct] _{train_tracks_structure_visual_appearance}	An assumption that fallen vegetation over train tracks obstructs the visual appearance of the train tracks' structure, potentially affecting maintenance and safety inspections.
119	{fallen_vegetation_over_train_tracks_structure}_[obstruct] _{powered_powerlines_cables_structure_visual_appearance}	Assuming that fallen vegetation over train tracks obstructs the visual appearance of powered power lines and cables, potentially affecting operational safety and maintenance.
120	{fallen_vegetation_over_train_tracks_structure}_[generate] _{leaves_&_pollen}	An assumption that fallen vegetation over train tracks generates leaves and pollen, potentially affecting visibility and maintenance efforts.
121	{police_force_existing_capability}_[monitor] _{roaming_adversarial_drone}	Assuming that the police force monitors roaming adversarial drones, potentially mitigating security risks.
122	{police_force_existing_capability}_[+support] _{safe_secure_train_track_zone}	An assumption that the police force supports ensuring a safe and secure train track zone, enhancing operational safety.
123	{police_force_existing_capability}_[+direct] _{police_officer_situational_awareness}	Assuming that the police force directs police officers to enhance situational awareness, improving response capabilities.
124	{police_officer_situational_awareness}_[+monitor] _{train_motion_dynamics}	An assumption that police officer situational awareness monitors approaching trains, ensuring safety and security protocols are followed.
125	{police_officer_situational_awareness}_[+support] _{safe_secure_train_track_zone}	Assuming that police officer situational awareness supports maintaining a safe and secure train track zone, enhancing operational safety.
126	{police_officer_situational_awareness}_[+support] _{safe_secure_police_officer}	An assumption that police officer situational awareness supports ensuring the safety and security of police officers themselves during operations.

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127	{roaming_adversarial_drone}_[-impede] _{birds_wasps_attack_expose_adversarial_drone}	Assuming that roaming adversarial drones impede train operations, potentially causing delays or disruptions.
128	{roaming_adversarial_drone}_[-avoid] _{police_officer_capture_adversarial_drone}	An assumption that roaming adversarial drones avoid capture by police officers, complicating security efforts.
129	{roaming_adversarial_drone}_[-fly_over] _{free_space_above_train_roof}	Assuming that roaming adversarial drones fly over the free space above train roofs, potentially posing security risks.
130	{stationary_train}_[support] _{roaming_adversarial_drone}	An assumption that a stationary train supports the presence of roaming adversarial drones unintentionally, possibly due to its stationary nature providing a perch or cover.
131	{stationary_train}_[impede] _{police_officer_capture_adversarial_drone}	Assuming that a stationary train inadvertently impedes police officers attempting to capture adversarial drones, potentially due to obstructed views or access.
132	{stationary_train}_[obstruct] _{police_officer_movements}	An assumption that a stationary train inadvertently obstructs police officer movements, possibly affecting response times or access to critical areas.
133	{roaming_adversarial_drone}_[appreciate] _{powered_powerlines_cables_structure_visual_appearance}	Assuming that a roaming adversarial drone appreciates the visual appearance of powered powerlines and cables structures, possibly for navigation or shelter goals.
134	{roaming_adversarial_drone}_[-hover] _{free_space_above_train_roof}	An assumption that a roaming adversarial drone hovers over the free space above train roofs, potentially posing a security risk or causing operational disruptions.

J.2 Extracted Predicted Situations for PAD analysis:

Salient Interaction Scenario	Typical Undesirable Scenarios	Extracted Predicted Situations	{entity1_aspect_verb_entity2_aspect}	Train prevent Adversarial Drone
h	{moving_train_aerodynamic_wake_destabilize_adversarial_drone_flight }_{train_emergency_manoeuvre_destabilize_adversarial_drone_control }_{destabilized_adversarial_drone_fail_complete_mission}	Infrastructure	obstruct	Adversarial Drone
	{trackside_fencing_entangle_adversarial_drone_flight }_{overhead_lines_block_adversarial_drone_path }_{trackside_signage_force_adversarial_drone_cras	Environment	impact	obstructs Adversarial Drone
	{_obstructed_adversarial_drone_abort_planned_mission}	Environment	impact	obstructs Adversarial Drone
sion	{strong_gusts_degrade_adversarial_drone_navigation }_{rain_conditions_degrade_adversarial_drone_sensors }_{thermal_turbulence_destabilize_adversarial_drone_flight }_{degraded_adversarial_drone_abort_mission }_{environment_conditions_cause_adversarial_drone_collision}	Passengers	ignore the	Adversarial drone
	{unaware_passengers_disregard_drone_warnings }_{mobile_passengers_enter_adversarial_drone_corridor }_{blocked_adversarial_drone_hover_abort_mis	Adversarial	Drone	disrupt Train
	{_hovering_adversarial_drone_collide_passengers}	Adversarial	Drone	disrupt Train
ransit	{low_altitude_adversarial_drone_trigger_detection_alarms }_{adversarial_drone_presence_cause_operator_braking }_{emergency_braking_interrupt_train_t	Infrastructure	fail	Train
	{_disrupted_train_service_delay_schedule}	Infrastructure	fail	Train
	{damaged_track_components_force_train_slowdown }_{displaced_track_sections_prevent_train_passage }_{failed_rail_infrastructure_halt_train_operation	Environment	impact	obstruct Train
s}	{fog_conditions_reduce_train_visibility }_{ice_conditions_reduce_train_braking }_{wet_rails_obstruct_train_traversal }_{degraded_visibility_prevent_safe_tra	Passengers	not	travel Train
	{_perceived_risk_cause_passenger_avoidance }_{widespread_avoidance_leave_train_underutilised }_{underutilised_train_service_undermine_transit_missi	Adversarial	Drone	collide Infrastructure
	{adversarial_drone_impact_sever_trackside_cables }_{adversarial_drone_collision_dislodge_trackside_fencing }_{damaged_infrastructure_create_track_h	Train	collide	Infrastructure
azards	{_degraded_infrastructure_compromise_safety_function}	Train	collide	Infrastructure
	{moving_train_strike_trackside_debris }_{train_collision_break_rail_components }_{train_impact_damage_signal_infrastructure }_{broken_infrastructure_ha	Environment	impact	degrades infrastructure
	{_lt_train_operations}	Environment	impact	degrades infrastructure
ity	{flooding_conditions_undermine_track_bed }_{erosion_processes_degrade_trackside_fencing }_{vegetation_overgrowth_compromise_infrastructure_integr	Passengers	degrade	Infrastructure
	{_nesting_birds_degrade_signal_infrastructure }_{wasp_hives_obstruct_infrastructure_components}	Passengers	degrade	Infrastructure
	{vandalism_actions_damage_trackside_fencing }_{littering_passengers_degrade_infrastructure_cleanliness }_{forcing_actions_break_security_gates }_{da	Adversarial	Drone	complicates Environment impact
maged_gates_compromise_containment_function}	{adversarial_drone_propwash_displace_loose_objects }_{adversarial_drone_debris_create_sensor_occlusions }_{displaced_dust_create_visibility_hazards	Train	complicates	Environment impact
	{_altered_conditions_complicate_sensor_detection}	Train	complicates	Environment impact
	{moving_train_wake_change_local_airflow }_{train_exhaust_create_heat_distortion }_{train_heat_worsen_drone_sensor_conditions }_{altered_airflow_degr	Infrastructure	complicates	Environment impact
ade_nearby_system_sensors}	{_temporary_barriers_redirect_wind_patterns }_{new_structures_create_sensor_reflections }_{redirected_water_produce_unexpected_microclimate }_{infra	Passengers	ignore	Environment impact
	{structure_glare_impair_drone_sensors}	Passengers	ignore	Environment impact

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{unaware_passengers_ignore_ice_hazards }_{mobile_passengers_enter_flooded_areas }_{ignoring_passengers_increase_incident_likelihood }_{unsafe_passenger_behaviour_obstruct_safe_transit}	Adversarial	Drone	annoy	Passengers
{adversarial_drone_noise_alarm_waiting_passengers }_{adversarial_drone_proximity_irritate_passengers }_{sudden_drone_movements_provoke_unsafe_reactions }_{alarmed_passengers_create_crowding_hazards}	Train	delay	Passengers	
{delayed_train_service_cause_passenger_crowding }_{train_delays_cause_missed_connections }_{stressed_passengers_degrade_safe_behaviour }_{crowded_platforms_compromise_transit_safety}	Infrastructure	prevent	Passengers	
{locked_gates_prevent_passenger_boarding }_{broken_access_points_prevent_passenger_alighting }_{barrier_infrastructure_obstruct_passenger_evacuation }_{prevented_evacuation_compromise_passenger_safety}	Environment	impact	annoy	Passengers
{adverse_weather_cause_passenger_distress }_{cold_conditions_provoke_unsafe_passenger_behaviours }_{heat_exposure_degrade_passenger_comfort }_{glare_conditions_impair_passenger_navigation}	Atypical Salient Interaction Scenario	Atypical scenarios:	Extracted	Predicted Situations
{entity1_aspect_verb_entity2_aspect} Train prevent Adversarial Drone	{cooperative_adversarial_drone_wait_near_wake_zones}			
{hovering_adversarial_drone_allow_moving_train_approach}		{moving_train_aerodynamic_wake_cause_drone_abort}		
{waiting_adversarial_drone_support_train_mission_prevention}	Infrastructure	obstruct	Drone	
{failing_adversarial_drone_support_trackside_fencing_obstruction}		{adversarial_drone_fail_avoid_overhead_lines}		
{ineffective_fencing_enable_adversarial_drone_strikes} {permissive_infrastructure_allow_hostile_drone_passage}	Environment impact obstructs	Adversarial Drone		
{hovering_adversarial_drone_allow_glare_conditions_degrade}		{adversarial_drone_remain_glare_prone_zones}		
{environment_conditions_cause_self_limiting_drone_failure} {cooperative_adversarial_drone_support_sensor_degradation}	Passengers ignore	Drone		
{signalling_adversarial_drone_enable_mobile_passengers_enter}		{benign_adversarial_drone_allow_passengers_enter_corridor}		
{mobile_passengers_prevent_adversarial_drone_clearance} {cooperative_adversarial_drone_support_passenger_interference}	Adversarial Drone disrupt	Train		
{negligent_train_allow_adversarial_drone_strike} {permissive_train_operations_create_collusive_hazards}	{tacit_train_permit_low_altitude_incursions}			
{complicit_train_support_adversarial_drone_attacks} {subverted_safety_protocols_enable_drone_strikes}	Infrastructure fail	Train		
{operating_train_allow_degraded_infrastructure_remain}		{train_operations_support_infrastructure_degradation}		
{paradoxical_train_prevent_infrastructure_safe_function} {degraded_infrastructure_enabled_by_train_tolerance}	Environment impact obstruct	Train		
{operating_train_allow_fog_conditions_reduce} {train_operations_worsen_visibility_conditions}	{fog_prone_schedule_support_environment_obstruction}			
{cooperative_train_enable_unsafe_transit_conditions}	Passengers do not travel	Train	{risk_signalling_train_allow_passenger_avoidance}	
{train_operations_enable_widespread_passenger_avoidance}			{perceived_train_risk_lead_service_underutilisation}	
{tolerant_train_signal_unintended_passenger_deterrence}	Adversarial Drone collide	Infrastructure	{exposed_infrastructure_allow_adversarial_drone_impact}	
{trackside_infrastructure_support_adversarial_drone_collision}			{permissive_infrastructure_facilitate_drone_damage}	
{paradoxical_infrastructure_enable_self_degradation}	Train collide	Infrastructure	{permissive_infrastructure_allow_moving_train_collision}	
{debris_laden_infrastructure_enable_train_strikes} {gapped_track_infrastructure_facilitate_train_impact}	{paradoxical_infrastructure_support_collision_events}			
Environment impact degrade	Infrastructure	{unmaintained_infrastructure_allow_flooding_conditions_erode}		
{lacking_infrastructure_support_environment_degradation}		{passive_infrastructure_enable_vegetation_overgrowth_compromise}		
{permissive_infrastructure_fail_maintain_integrity}	Passengers degrade	Infrastructure	{accessible_infrastructure_allow_vandalism_actions_damage}	
{easy_access_gates_support_passenger_vandalism}			{permissive_infrastructure_enable_littering_passengers_degrade}	
{paradoxical_infrastructure_facilitate_self_damage}	Adversarial	Drone	complicates	Environment impact
{amplifying_environment_conditions_allow_adversarial_drone_propwash}		{environment_conditions_support_adversarial_drone_sensor_occlusion}		

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{reflective_environment_enable_drone_sensing_degradation} {paradoxical_environment_worsen_adversarial_drone_operations} Train complicates Environment
 impact {trapping_environment_conditions_allow_moving_train_worsen} {environment_conditions_support_train_heat_distortion}
 {local_environment_enable_train_turbulence_intensify} {amplifying_environment_worsen_drone_sensor_disruption} Infrastructure complicates Environment
 impact {redirecting_environment_conditions_allow_temporary_barriers_redirect} {environment_flow_support_new_structures_glare}
 {amplifying_environment_enable_infrastructure_microclimate_creation} {nesting_birds_deter_adversarial_drone_intrusions}
 {wasp_hives_provide_natural_drone_deterrence} Passengers ignore Environment impact {masking_environment_conditions_allow_unaware_passengers_enter}
 {environment_hazards_support_mobile_passengers_ignore} {concealing_ice_conditions_enable_risky_passenger_behaviour}
 {paradoxical_environment_facilitate_unsafe_passenger_actions} Adversarial Drone annoy Passengers
 {tolerant_passengers_allow_adversarial_drone_noise_continue} {accepting_passengers_support_adversarial_drone_proximity}
 {passive_passengers_enable_continued_drone_annoyance} {permissive_passengers_reinforce_drone_operations} Train delay Passengers
 {accepting_passengers_allow_delayed_train_service_continue} {tolerant_passengers_support_train_schedule_slippage}
 {waiting_passengers_enable_extended_train_delays} {passive_passengers_reinforce_delay_patterns} Infrastructure prevent Passengers
 {passive_passengers_allow_locked_gates_restrict} {accepting_passengers_support_barrier_infrastructure_prevention}
 {non_seeking_passengers_enable_boarding_restrictions} {permissive_passengers_reinforce_access_barriers} Environment impact annoy Passengers
 {acclimatising_passengers_allow_adverse_weather_remain} {tolerant_passengers_support_cold_conditions_persist}
 {accepting_passengers_enable_persistent_environment_discomfort} {passive_passengers_reinforce_uncomfortable_conditions} Step 1: Select observed
 outcome behaviour: Unsafe problematic activities step 2: observed system (obs) Step 3: Inquired interaction step 4: supra source primary purpose
 moving_train 1) |{unsafe_moving_train}_[-approach]_{unsafe_train_tracks_zone_structure}| Note that we used (-) because, currently, the moving train worsens the
 unsafe train track situation. primary purpose: transport passengers or goods along the designated train tracks. step 5: auxiliary influence interaction
 step 6: auxiliary control interaction step 7: auxiliary appreciation interaction influence goal: 1.
 |{unsafe_moving_train}_[maintains]_{safe_passengers_and_goods}| influence actions: 1.
 |{unsafe_moving_train}_[inform]_{train_driver_situational_awareness}| 2. |{unsafe_moving_train}_[prevent]_{unsafe_unsecured_passengers_and_goods}|
 control goals: 1. |{unsafe_moving_train}_[control]_{onboard_train_safety_systems}| 2. |{unsafe_moving_train}_[control]_{alert_systems}| control
 actions: for {onboard_train_safety_systems} (preventing unsafe passengers/goods): 1. |{unsafe_moving_train}_[activate]_{onboard_train_safety_systems}| for
 {alert_systems} (informing the train driver): 2. |{unsafe_moving_train}_[trigger]_{alert_systems}| 3.
 |{unsafe_moving_train}_[alert]_{train_driver_situational_awareness}| appreciation goal: 1.
 |{unsafe_moving_train}_[appreciate]_{track_zone_monitoring_systems}| appreciative actions: 1.
 |{unsafe_moving_train}_[initiate_collection]_{track_zone_monitoring_systems_data}| 2.
 |{unsafe_moving_train}_[acknowledge]_{timely_track_zone_monitoring_systems_data}| 3.
 |{unsafe_moving_train}_[acknowledge]_{accurate_track_zone_monitoring_systems_data}| 4.
 |{unsafe_moving_train}_[highlight]_{track_zone_monitoring_systems_data_gaps_or_inconsistencies}| step 2: observed system (obs) Step 3: Inquired
 interaction step 4: supra source primary purpose moving_train, roaming_adversarial_drone
 2) |{unsafe_moving_train}_[+approach]_{roaming_adversarial_drone}| Note that we used (-) because, currently, the moving train improves the chances that
 the adversarial drone will achieve its goals. primary purpose: transport passengers or goods along the designated train tracks. step 5: auxiliary influence
 interaction step 6: auxiliary control interaction step 7: auxiliary appreciation interaction influence goal:
 |{unsafe_moving_train}_[influence]_{roaming_adversarial_drones}| influence actions: 1. communication (ground-based security systems): 1.1

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{unsafe_moving_train}_[inform]_{available_ground_based_security_systems}}	1.2
{unsafe_moving_train}_[communicate]_{roaming_adversarial_drones_location_data}}	1.3
{unsafe_moving_train}_[communicate]_{roaming_adversarial_drones_information}} 2. coordination (emergency response teams):	2.1
{unsafe_moving_train}_[trigger_communication]_{unaware_emergency_response_teams}} control goal 1:	1.
{unsafe_moving_train}_[control]_{available_ground_based_security_systems}}	2.
{unsafe_moving_train}_[control]_{roaming_adversarial_drones_location_data}}	3.
{unsafe_moving_train}_[control]_{roaming_adversarial_drones_information}} control actions 1: for communication interface:	1.
{unsafe_moving_train}_[establish_connection]_{available_ground_based_security_systems}}	2.
{unsafe_moving_train}_[transmit_data_packets]_{onboard_train_communication_interface}} for data processing unit:	3.
{unsafe_moving_train}_[filter]_{roaming_adversarial_drones_size_data}} 4. {{unsafe_moving_train}_[filter]_{roaming_adversarial_drones_location_data}}	5.
{unsafe_moving_train}_[filter]_{roaming_adversarial_drones_flying_behaviour_data}}	6.
{unsafe_moving_train}_[prioritize]_{roaming_adversarial_drones_critical_data}}	7.
{unsafe_moving_train}_[format]_{roaming_adversarial_drones_information_transmitted_data}} control goal 2:	4.
{unsafe_moving_train}_[trigger_communication]_{unaware_emergency_response_teams}} control actions 2: for communication interface:	5.
{unsafe_moving_train}_[establish_connection]_{designated_emergency_response_communication_system}}	6.
{unsafe_moving_train}_[transmit]_{onboard_train_communication_interface}}{emergency_notification}} 1.	
{unsafe_moving_train}_[appreciate]_{available_ground_based_security_systems}}	2.
{unsafe_moving_train}_[appreciate]_{reliable_railways_traffic_control}}	3.
{unsafe_moving_train}_[appreciate]_{accurate_track_zone_monitoring_systems}} appreciative actions:	4.
{unsafe_moving_train}_[monitor]_{available_ground_based_security_systems}} 5. {{unsafe_moving_train}_[monitor]_{communication_network_connectivity}}	6.
{unsafe_moving_train}_[establish]_{alternative_communication_channels}} 7. {{unsafe_moving_train}_[verify]_{track_zone_monitoring_systems_data_plausibility}}	
8. {{unsafe_moving_train}_[confirms]_{ground_based_security_systems_received_data}}	9.
{unsafe_moving_train}_[assess]_{disruptive_adversarial_drones_threat_severity}} step 8: Predicted outcome emergence contributing situations (with repetition)	
Step 1: Select observed outcome behaviour: Unsafe problematic activities Step 2: observed system (obs) Step 3: Inquired interaction step 4: supra source	
primary purpose roaming_adversarial_drone {{roaming_adversarial_drone}_[-approach]_{unsafe_moving_train}} primary purpose: disrupt or gather intelligence	
within the train tracks zone by evading detection and obstacles. Step 5: auxiliary influence interaction Step 6: auxiliary control interaction step 7:	
auxiliary appreciation interaction influence goals: 1. {{roaming_adversarial_drone}_[influence]_{train_driver_situational_awareness}}	
{roaming_adversarial_drone}_[influence]_{onboard_train_safety_systems}} influence actions:	
{roaming_adversarial_drone}_[maintain]_{low_altitude_roaming_adversarial_drone}}	
{roaming_adversarial_drone}_[employ]_{roaming_adversarial_drone_camouflage}}	
{roaming_adversarial_drone}_[suppress]_{roaming_adversarial_drone_noise_emissions}} 6. {{roaming_adversarial_drone}_[exploit]_{moving_train_blind_spots}}	
control goals: 1. {{roaming_adversarial_drone}_[control]_{altitude_control_systems}}	
{roaming_adversarial_drone}_[control]_{camouflage_mechanisms}}	
{roaming_adversarial_drone}_[control]_{roaming_adversarial_drone_noise_suppression_systems}}	
{roaming_adversarial_drone}_[control]_{roaming_adversarial_drone_navigation_systems}} control actions:	
{roaming_adversarial_drone}_[adjust]_{altitude_parameters}} 2. {{roaming_adversarial_drone}_[activate]_{camouflage_mechanisms}}	

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{{roaming_adversarial_drone}_[reduce]_{{roaming_adversarial_drone_noise_output}}}	4.
{{roaming_adversarial_drone}_[optimize]_{{path_planning_algorithms}}}	5.
{{roaming_adversarial_drone}_[monitor]_{{bad_weather}}}	6.
{{roaming_adversarial_drone}_[adapt]_{{adversarial_drone_response_strategies}}}	1.
appreciation goals:	
{{roaming_adversarial_drone}_[appreciate]_{{bad_weather}}}	2.
{{roaming_adversarial_drone}_[appreciate]_{{train_schedule_timing}}}	3.
{{roaming_adversarial_drone}_[appreciate]_{{train_tracks_maintenance_activities}}}	4.
{{roaming_adversarial_drone}_[appreciate]_{{track_zone_obstacles}}}	1.
appreciative actions:	
1. {{roaming_adversarial_drone}_[monitor]_{{bad_weather}}}	2.
{{roaming_adversarial_drone}_[synchronize]_{{train_schedule_timing}}}	3.
{{roaming_adversarial_drone}_[scan]_{{train_tracks_maintenance_activities}}}	4.
{{roaming_adversarial_drone}_[detect]_{{track_zone_obstacles}}}	5.
{{roaming_adversarial_drone}_[adapt]_{{flight_path_based_on_weather}}}	6.
{{roaming_adversarial_drone}_[plan]_{{routes_around_maintenance_activities}}}	7.
{{roaming_adversarial_drone}_[navigate]_{{adversarial_drone_dangerous_navigation_around_obstacles}}}	step 8: Predicted outcome emergence contributing situations (with repetition)
Step 2: observed system (obs)	Step 3: Inquired interaction
Step 4: supra source	primary purpose
roaming_adversarial_drone	{{roaming_adversarial_drone}_[+manage]_{{invisible_roaming_adversarial_drone_visual_appearance}}}
The roaming adversarial drone manages (controls or influences) its own visual appearance to achieve invisibility or low observability, thereby evading detection by visual perception systems (e.g., Eagle Drone cameras, track zone monitoring systems).	primary purpose: disrupt or gather intelligence within the train tracks zone by evading detection and obstacles.
Step 5: auxiliary influence interaction	Step 6: auxiliary control interaction
step 7: auxiliary appreciation interaction	Influence Goal:
1. {{roaming_adversarial_drone}_[influence]_{{inadequate_eagle_drone_perception_reliability}}}	2.
{{roaming_adversarial_drone}_[influence]_{{inadequate_track_zone_monitoring_systems_detection_capability}}}	3.
{{roaming_adversarial_drone}_[influence]_{{unaware_train_driver_situational_awareness}}}	Influence Actions:
1. For	
{{inadequate_eagle_drone_perception_reliability}}:	1.1
{{roaming_adversarial_drone}_[reduce]_{{detectable_visual_signature}}}	1.2
{{roaming_adversarial_drone}_[exploit]_{{low_contrast_environmental_conditions}}}	1.3
{{roaming_adversarial_drone}_[mimic]_{{benign_environmental_object_appearance}}}	2.
For	
{{inadequate_track_zone_monitoring_systems_detection_capability}}:	2.1
{{roaming_adversarial_drone}_[blend]_{{ambient_background_visual_texture}}}	2.2
{{roaming_adversarial_drone}_[obscure]_{{characteristic_drone_shape_features}}}	3.
For	
{{unaware_train_driver_situational_awareness}}:	3.1
{{roaming_adversarial_drone}_[delay]_{{timely_threat_notification}}}	3.2
{{roaming_adversarial_drone}_[degrade]_{{reliable_security_alert_information}}}	Control
Goal 1: 1.	
{{roaming_adversarial_drone}_[control]_{{adaptive_camouflage_system}}}	2.
{{roaming_adversarial_drone}_[control]_{{low_observability_coating_material}}}	3.
{{roaming_adversarial_drone}_[control]_{{dynamic_surface_pattern_generator}}}	Control Actions
1: For	
{{adaptive_camouflage_system}}:	1.
{{roaming_adversarial_drone}_[activate]_{{adaptive_camouflage_system}}}	2.
{{roaming_adversarial_drone}_[adjust]_{{real_time_camouflage_pattern}}}	3.
{{roaming_adversarial_drone}_[synchronize]_{{background_environmental_visual_data}}}	For
{{low_observability_coating_material}}:	4.
{{roaming_adversarial_drone}_[apply]_{{low_reflectivity_coating_material}}}	5.
{{roaming_adversarial_drone}_[minimize]_{{specular_reflection_intensity}}}	For
{{dynamic_surface_pattern_generator}}:	6.
{{roaming_adversarial_drone}_[generate]_{{adaptive_visual_pattern}}}	7.
{{roaming_adversarial_drone}_[modulate]_{{surface_color_distribution}}}	8.
{{roaming_adversarial_drone}_[match]_{{ambient_lighting_conditions}}}	Control
Goal	
2:	4.
{{roaming_adversarial_drone}_[control]_{{flight_trajectory_profile}}}	5.
{{roaming_adversarial_drone}_[control]_{{timing_intrusion_window}}}	Control
Actions 2: For	
{{flight_trajectory_profile}}:	9.
{{roaming_adversarial_drone}_[select]_{{low_visibility_flight_path}}}	10.
{{roaming_adversarial_drone}_[avoid]_{{high_illumination_zones}}}	11.
{{roaming_adversarial_drone}_[exploit]_{{visual_occlusion_opportunities}}}	For

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{timing_intrusion_window}: 12. {{roaming_adversarial_drone}_[calculate]_{optimal_intrusion_timing}}		13.
{{roaming_adversarial_drone}_[delay]_{intrusion_until_low_detection_probability}}		14.
{{roaming_adversarial_drone}_[execute]_{time_coordinated_penetration}} Appreciation		Goal: 1.
{{roaming_adversarial_drone}_[appreciate]_{current_environmental_lighting_conditions}}		2.
{{roaming_adversarial_drone}_[appreciate]_{active_eagle_drone_patrol_pattern}}		3.
{{roaming_adversarial_drone}_[appreciate]_{available_visual_occlusion_features}} Appreciative Actions: For {current_environmental_lighting_conditions}: 1.		
{{roaming_adversarial_drone}_[monitor]_{ambient_illumination_intensity}} 2. {{roaming_adversarial_drone}_[detect]_{shadow_zone_locations}}		3.
{{roaming_adversarial_drone}_[recognize]_{low_contrast_background_textures}} For {active_eagle_drone_patrol_pattern}: 4.		
{{roaming_adversarial_drone}_[observe]_{eagle_drone_flight_trajectory}} 5. {{roaming_adversarial_drone}_[predict]_{eagle_drone_sensor_coverage_gaps}}		
6.	{{roaming_adversarial_drone}_[identify]_{predictable_patrol_schedule}} For {available_visual_occlusion_features}: 7.	
	{{roaming_adversarial_drone}_[scan]_{trackside_vegetation_density}} 8. {{roaming_adversarial_drone}_[locate]_{infrastructure_shadow_zones}}	9.
	{{roaming_adversarial_drone}_[assess]_{fence_structure_occlusion_potential}}	10.
	{{roaming_adversarial_drone}_[verify]_{adequate_concealment_coverage}} step 8: Predicted outcome emergence contributing situations (with repetition)	
For simplicity, we will stop the AIC analysis here, and we focus on the high-level unsafe problematic interactions: Unsafe_Scenario1.1 = {{unsafe_moving_train}_[-approach]_{unsafe_train_tracks_zone_structure}}, {{unsafe_moving_train}_[ignore]_{unaware_train_driver_situational_awareness}}, {{unsafe_moving_train}_[-lead]_{unsafe_unsecured_passengers_and_goods}} unsafe_scenario1.4 ={{unsafe_moving_train}_[agitates]_{random_vegetation_motion}},		
{{unsafe_moving_train}_[induces]_{traveling_train_induced_air_turbulence_and_vibration}}, {{unsafe_moving_train}_[pass_through]_{fence_to_train_open_space}},		
{{unsafe_moving_train}_[impede]_{any_trained_computer_vision_agent_perception_capability}}, unsafe_scenario1.5		
= {{unsafe_moving_train}_[manage]_{train_electric_power_supply}}, {{unsafe_moving_train}_[deploy]_{extended_train_pantograph}},		
{{unsafe_moving_train}_[ignore]_{free_space_above_train_roof}},		
{{extended_train_pantograph}_[impede]_{any_trained_computer_vision_agent_perception_capability}},		
{{extended_train_pantograph}_[trigger]_{powered_powerlines_cables_structure_sparks}},		
{{extended_train_pantograph}_[chafe_&_wobble]_{powered_powerlines_cables_structure_visual_appearance}} unsafe_scenario2.1		
{{roaming_adversarial_drone}_[+manage]_{invisible_roaming_adversarial_drone_visual_appearance}}, {{roaming_adversarial_drone}_[-lead]_{train_derailment}},		
{{roaming_adversarial_drone}_[-navigate_around]_{train_driver_situational_awareness}}, unsafe_scenario2.2		
{{roaming_adversarial_drone_perception_capability}_[-trace]_{visible_train_tracks_appearance}}, {{roaming_adversarial_drone_perception_capability}_[-lead]_{train_derailment}}, {{roaming_adversarial_drone_perception_capability}_[+govern]_{roaming_adversarial_drone_flight_dynamics}} unsafe_scenario2.3		
(unsafe_scenario2.2 {{roaming_adversarial_drone}_[-detect]_{visible_train_tracks_zone_fence_structure}}, {{roaming_adversarial_drone}_[-lead]_{train_derailment}}, {{roaming_adversarial_drone}_[+govern]_{roaming_adversarial_drone_flight_dynamics}}, {{roaming_adversarial_drone}_[-approach]_{train_tracks_zone_fence_structure}} .unsafe_scenario2.4 {{roaming_adversarial_drone}_[+govern]_{roaming_adversarial_drone_flight_dynamics}},		
{{roaming_adversarial_drone}_[-recognise]_{visible_vegetation_visual_appearance}}, {{roaming_adversarial_drone}_[-lead]_{train_derailment}},		
{{roaming_adversarial_drone}_[-avoid]_{visible_vegetation_structure}} unsafe_scenario2.6 {{roaming_adversarial_drone}_[-recognise]_{powered_powerlines_cables_structure_visual_appearance}}, {{roaming_adversarial_drone}_[-lead]_{train_derailment}},		
{{roaming_adversarial_drone}_[-avoid]_{powered_powerlines_cables_structure_visual_appearance}},		
{{roaming_adversarial_drone}_[+govern]_{roaming_adversarial_drone_flight_dynamics}} .unsafe_scenario2.7		
{{roaming_adversarial_drone}_[ignore]_{powered_powerlines_cables_emf}}, {{roaming_adversarial_drone}_[-		

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recognise_{powered_powerlines_cables_structure_visual_appearance}},	{roaming_adversarial_drone}_{-collide}_{powered_powerlines_cables_structure}},	
{roaming_adversarial_drone}_{-lead}_{train_derailment}},		
{roaming_adversarial_drone}_{+govern}_{roaming_adversarial_drone_flight_dynamics}}		.unsafe_scenario3.2
{unsafe_train_tracks_zone_structure}_{ignore}_{visibility_obstructive_objects}},		
{unsafe_train_tracks_zone_structure}_{lead}_{adversarial_drone_collision_with_train}},		
{unsafe_train_tracks_zone_structure}_{guide}_{roaming_adversarial_drone_perception_capability}}		unsafe_scenario3.4
{unsafe_train_tracks_zone_structure}_{ignore}_{train_tracks_structure_visual_appearance}},		
{unsafe_train_tracks_zone_structure}_{impede}_{any_trained_computer_vision_agent_perception_capability}},		
{unsafe_train_tracks_zone_structure}_{complicates}_{vegetation_complex_shapes_sizes_visual_appearance}},		
{unsafe_train_tracks_zone_structure}_{complicates}_{visible_train_tracks_zone_fence_structure_visual_appearance}}		.unsafe_scenario3.5
{unsafe_train_tracks_zone_structure}_{ignore}_{train_tracks_structure_visual_appearance}},		
{unsafe_train_tracks_zone_structure}_{impede}_{any_trained_computer_vision_agent_perception_capability}},		
{unsafe_train_tracks_zone_structure}_{complicates}_{visible_powered_powerlines_cables_structure_visual_appearance}}		unsafe_scenario4.2
{train_tracks_zone_fence_structure}_{ignore}_{visible_train_tracks_zone_fence_structure_visual_appearance}},		
{train_tracks_zone_fence_structure}_{guide}_{roaming_adversarial_drone_perception_capability}},		
{train_tracks_zone_fence_structure}_{demarcate}_{train_tracks_structure_visual_appearance}}		unsafe_scenario4.4
{vegetation_complexity_growth}_{follow}_{train_tracks_zone_fence_structure}},		
{vegetation_complexity_growth}_{impede}_{any_trained_computer_vision_agent_perception_capability}},		
{vegetation_complexity_growth}_{generate}_{vegetation_complex_shapes_sizes_visual_appearance}}		unsafe_scenario4.5
{train_tracks_zone_fence_structure}_{ignore}_{visible_powered_powerlines_cables_structure_visual_appearance}},		
{train_tracks_zone_fence_structure}_{impede}_{any_trained_computer_vision_agent_perception_capability}},		
{train_tracks_zone_fence_structure}_{complicates}_{visible_powered_powerlines_cables_structure_visual_appearance}}		unsafe_scenario5.1
{vegetation_complexity_growth}_{adapt}_{train_track_zone_structure}},		
{vegetation_complexity_growth}_{complicate}_{unaware_train_driver_situational_awareness}},		
{vegetation_complexity_growth}_{generate}_{vegetation_complex_shapes_sizes_visual_appearance}}		unsafe_scenario5.3
{fallen_vegetation_over_train_tracks_structure}_{follow}_{passing_train_season_&_wind}},		
{fallen_vegetation_over_train_tracks_structure}_{impede}_{any_trained_computer_vision_agent_perception_capability}},		
{fallen_vegetation_over_train_tracks_structure}_{obstruct}_{train_tracks_structure_visual_appearance}}covered	in	4.4 unsafe_scenario5.5
{fallen_vegetation_over_train_tracks_structure}_{follow}_{passing_train_season_&_wind}},		
{fallen_vegetation_over_train_tracks_structure}_{impede}_{any_trained_computer_vision_agent_perception_capability}},		
{fallen_vegetation_over_train_tracks_structure}_{obstruct}_{powered_powerlines_cables_structure_visual_appearance}},		
{fallen_vegetation_over_train_tracks_structure}_{generate}_{leaves_&_pollen}}covered	in 3.5covered in 4.5covered in 5.5	unsafe_scenario7.1
{police_force_existing_capability}_{-monitor}_{roaming_adversarial_drone}},	{police_force_existing_capability}_{+support}_{safe_secure_train_track_zone}},	
{police_force_existing_capability}_{+direct}_{police_officer_situational_awareness}}		unsafe_scenario7.2
{police_officer_situational_awareness}_{+monitor}_{train_motion_dynamics}},		
{police_officer_situational_awareness}_{+support}_{safe_secure_train_track_zone}},		

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{{police_officer_situational_awareness}}_ [+support]_ {safe_secure_police_officer}}      unsafe_scenario8.1      {{roaming_adversarial_drone}}_ [-
avoid]_ {police_officer_situational_awareness}},      {{roaming_adversarial_drone}}_ [-instigate]_ {birds_wasps_attack_expose_adversarial_drone}},
{{roaming_adversarial_drone}}_ [-avoid]_ {police_officer_capture_adversarial_drone}}      unsafe_scenario8.2      {{roaming_adversarial_drone}}_ [-
avoid]_ {police_officer_situational_awareness}},      {{roaming_adversarial_drone}}_ [-impede]_ {police_officer_capture_adversarial_drone}},
{{roaming_adversarial_drone}}_ [-fly_over]_ {free_space_above_train_roof}}      unsafe_scenario9.1      {{stationary_train}}_ [support]_ {roaming_adversarial_drone}},
{{stationary_train}}_ [impede]_ {police_officer_capture_adversarial_drone}},      {{stationary_train}}_ [obstruct]_ {police_officer_movements}}      unsafe_scenario9.2
{{roaming_adversarial_drone}}_ [appreciate]_ {powered_powerlines_cables_structure_visual_appearance}},      {{roaming_adversarial_drone}}_ [-
impede]_ {police_officer_capture_adversarial_drone}},      {{roaming_adversarial_drone}}_ [-hover]_ {free_space_above_train_roof}}| below is the list of the remaining
discovered factors A useful python script to help with capturing situations from the interactions analysis can be used:
```